VIT

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

School of Computer Science Engineering & Information Systems (SCORE)

Lab Assessment 2

Name: Sumedha Challagundla

Registration number: 23MIS0175

Course title: Agile Development Process and Devops Lab

Slot: L57+L58

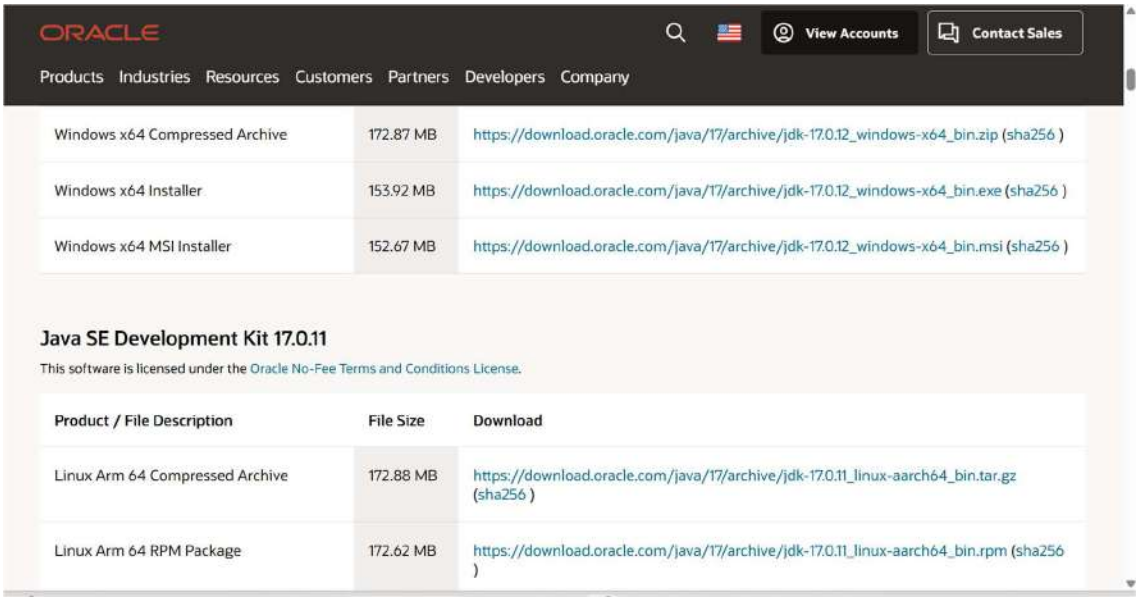
Course code: ISWE406P

Faculty name: Dr. Ramalingam M

9th jan, 2026 Lab

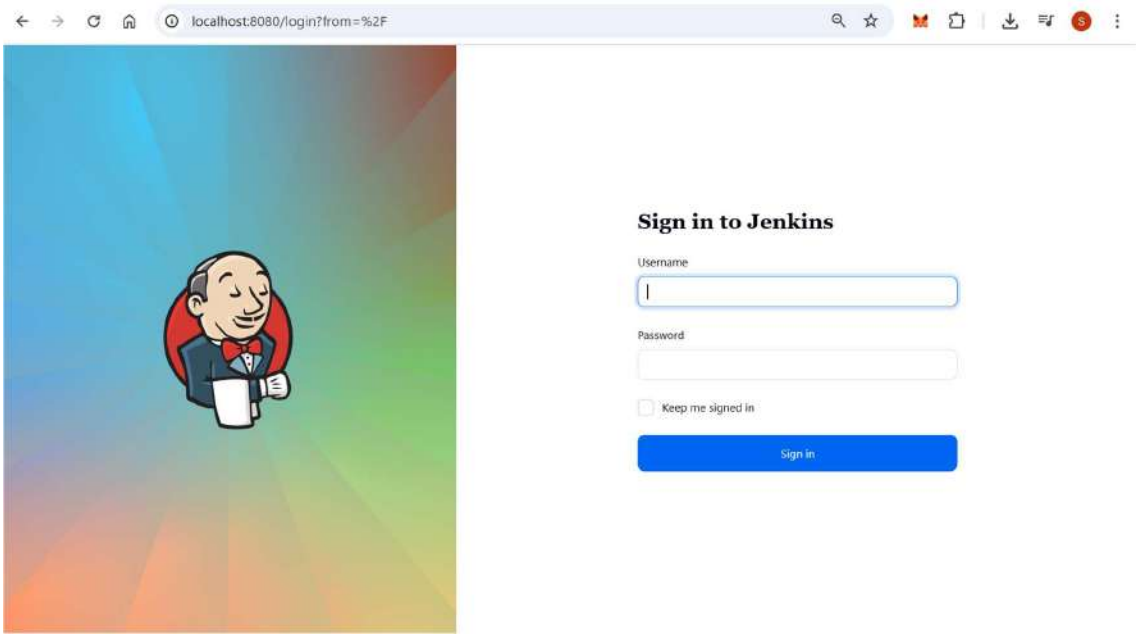
Installation of JAVA

JAVA 17-21 versions are compatible with Jenkins

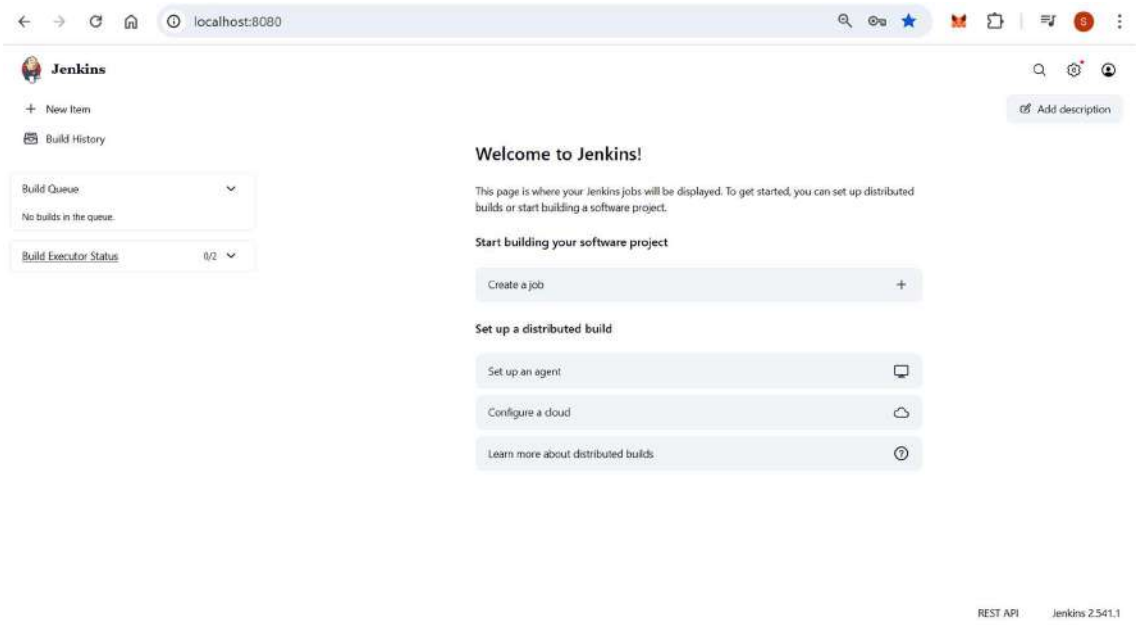


Installation of Jenkins

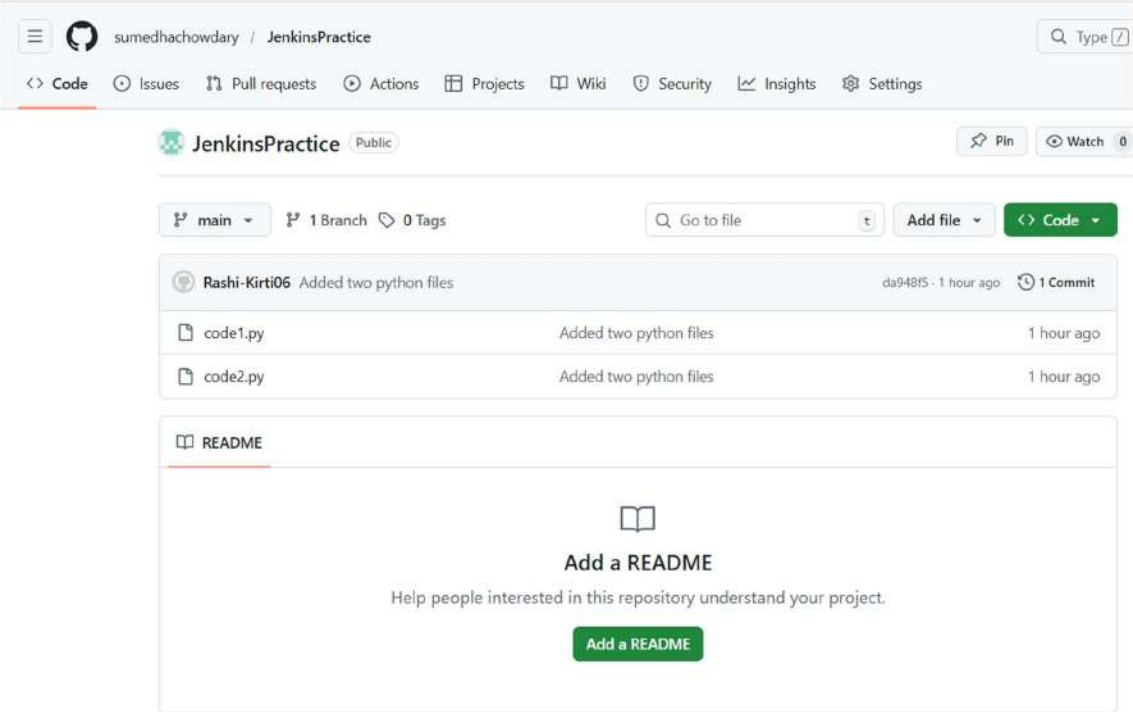
Sign in to Jenkins with registered credentials



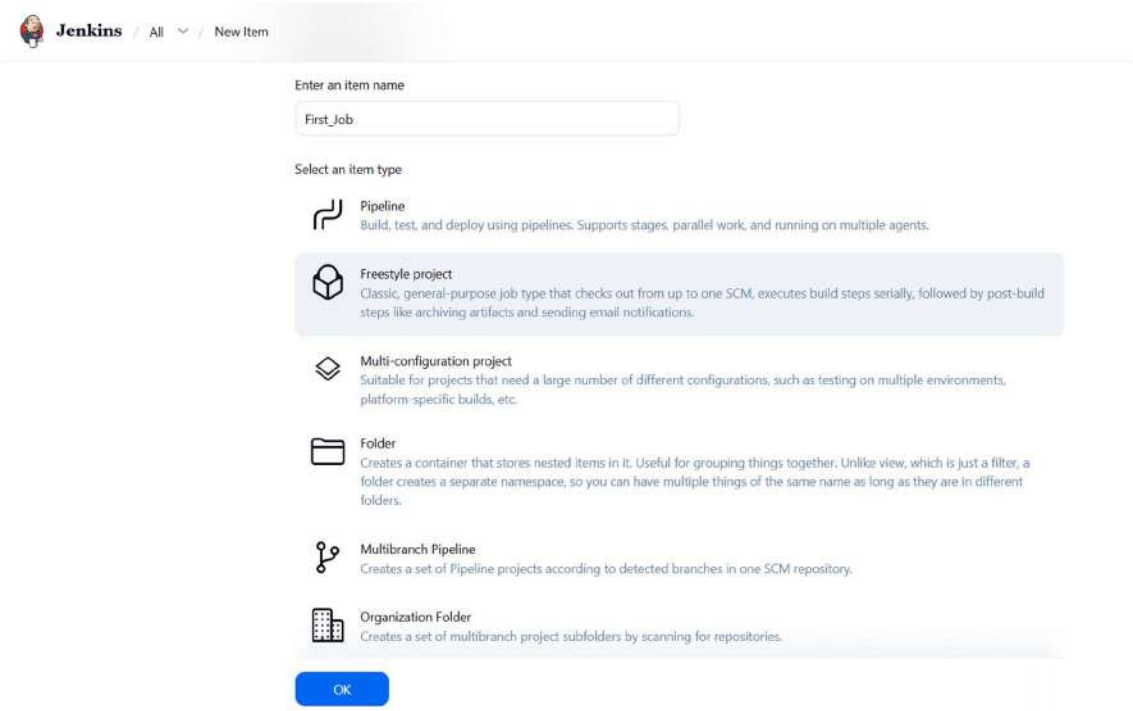
Running on port number 8080



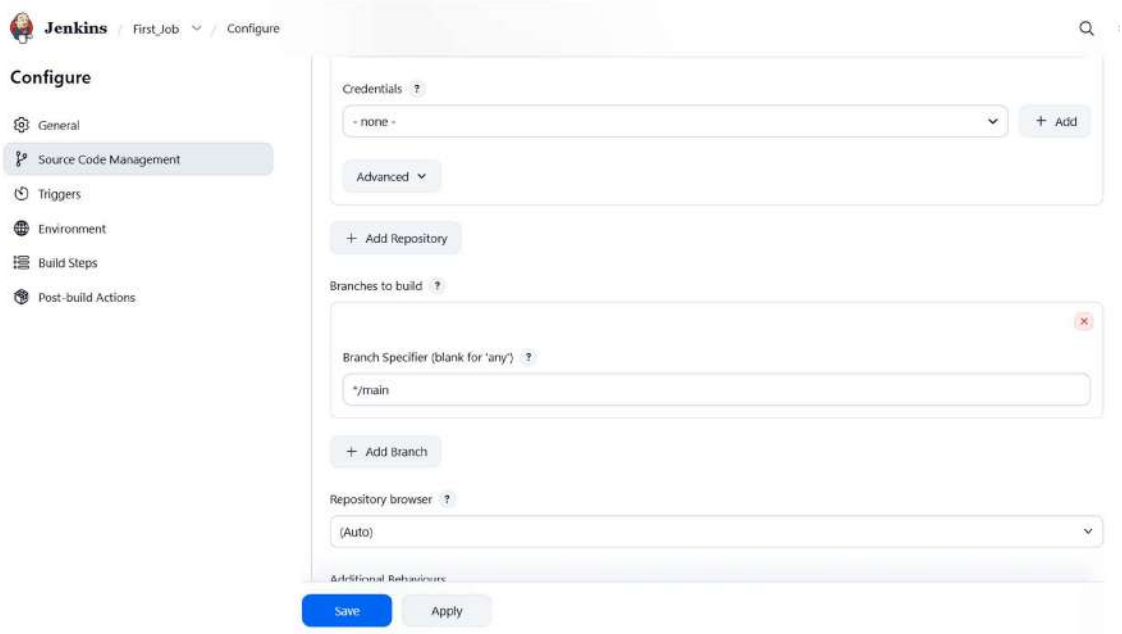
Git repository which will be used throughout



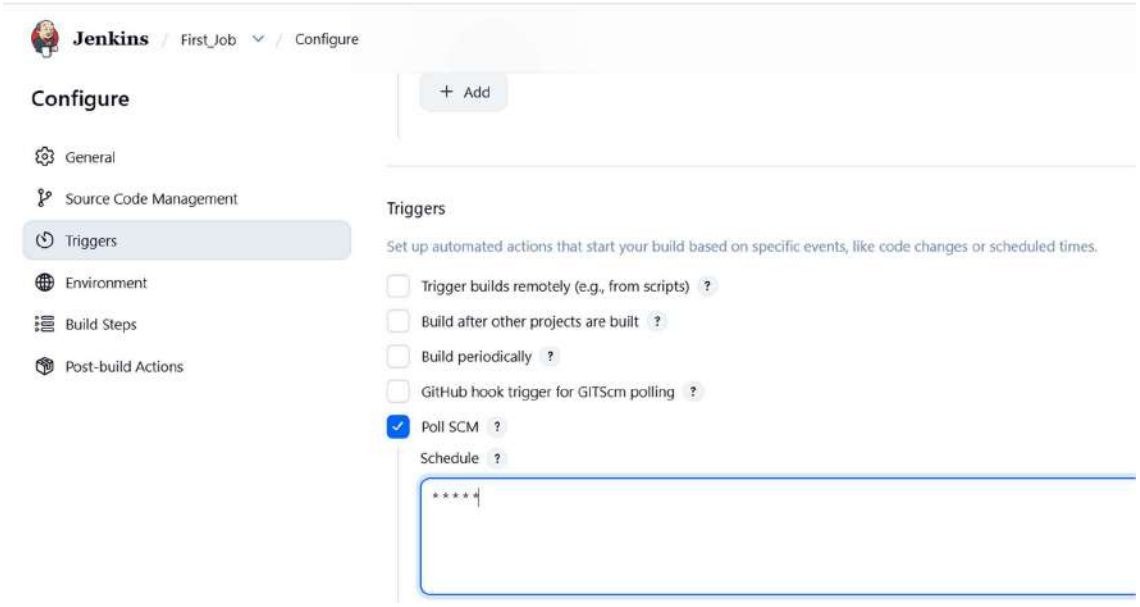
1.Creating a new job(Freestyle project)



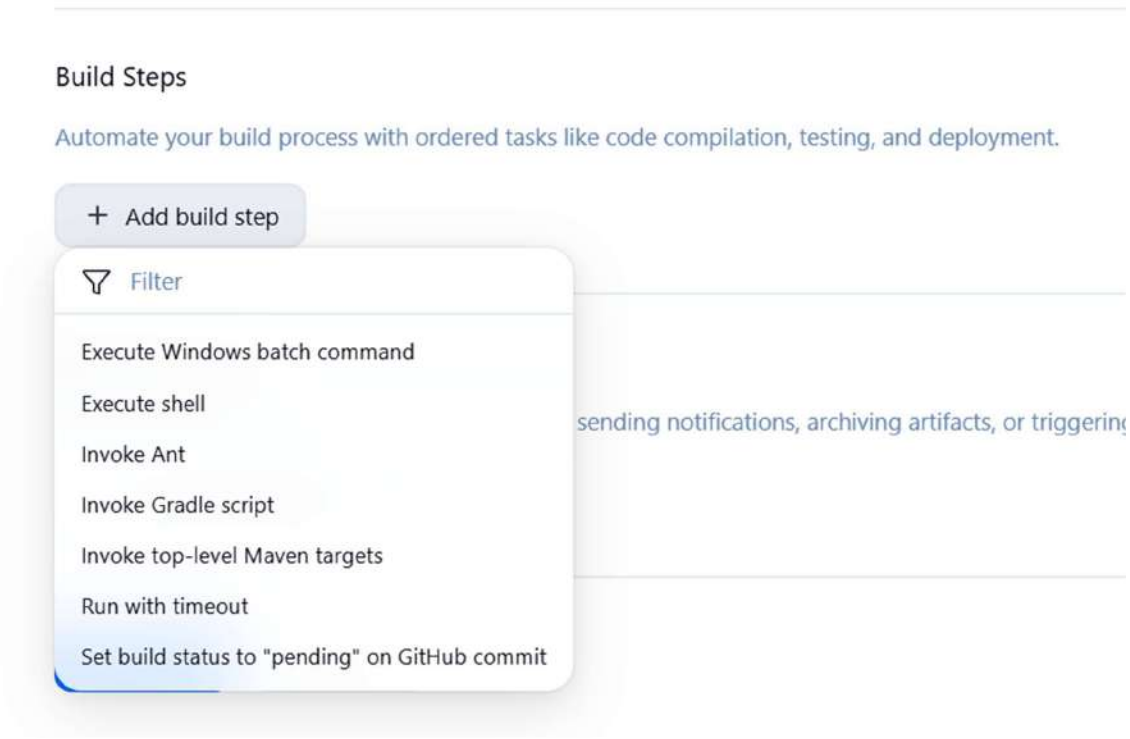
Add repository name and change branch specifier to */main



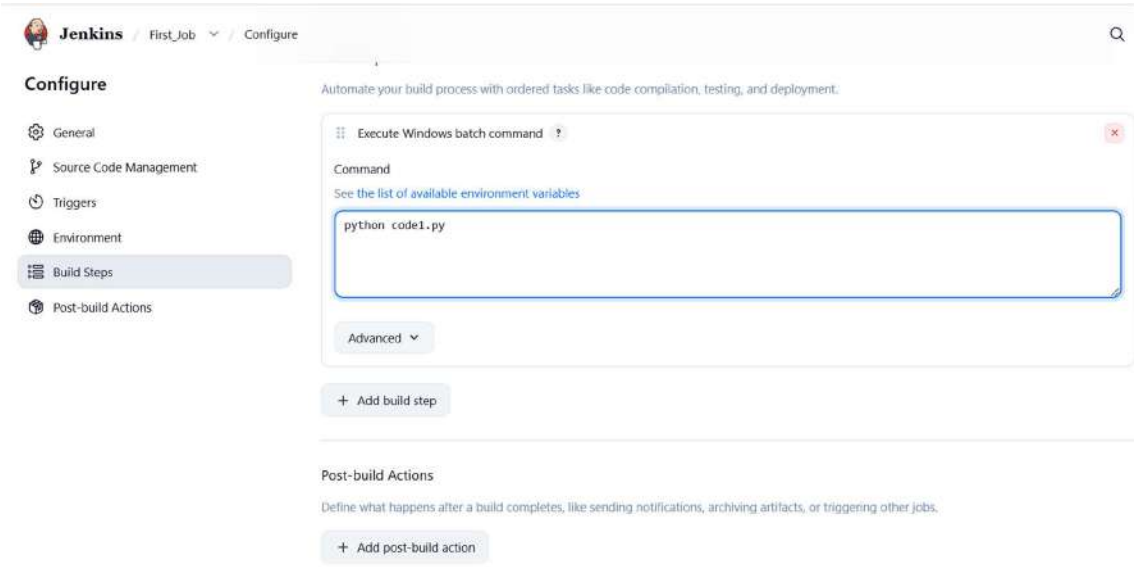
poll SCM Schedule trigger- * * * * *



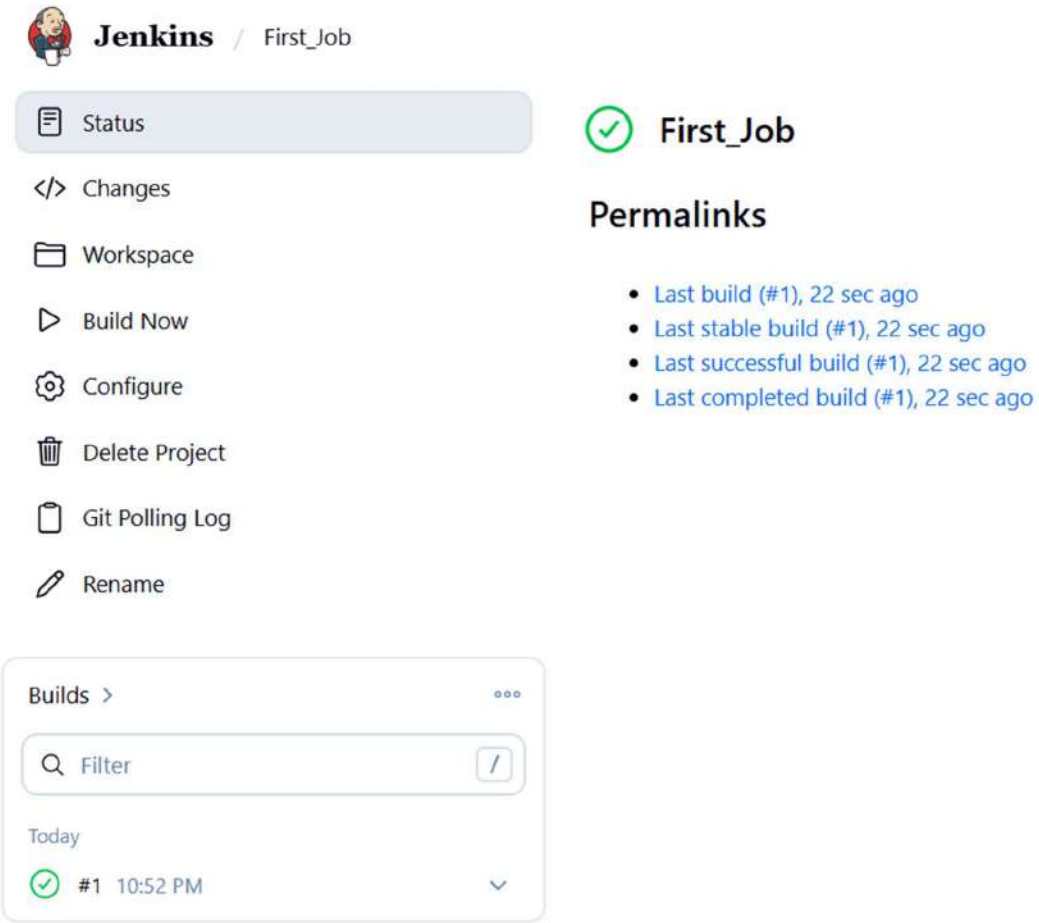
Build Steps-Execute batch command



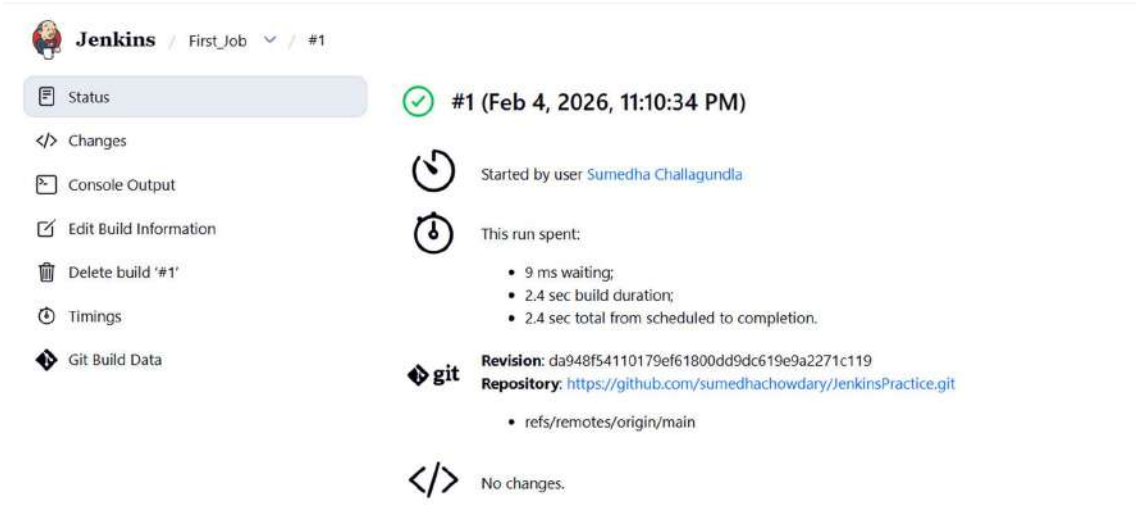
Add file name



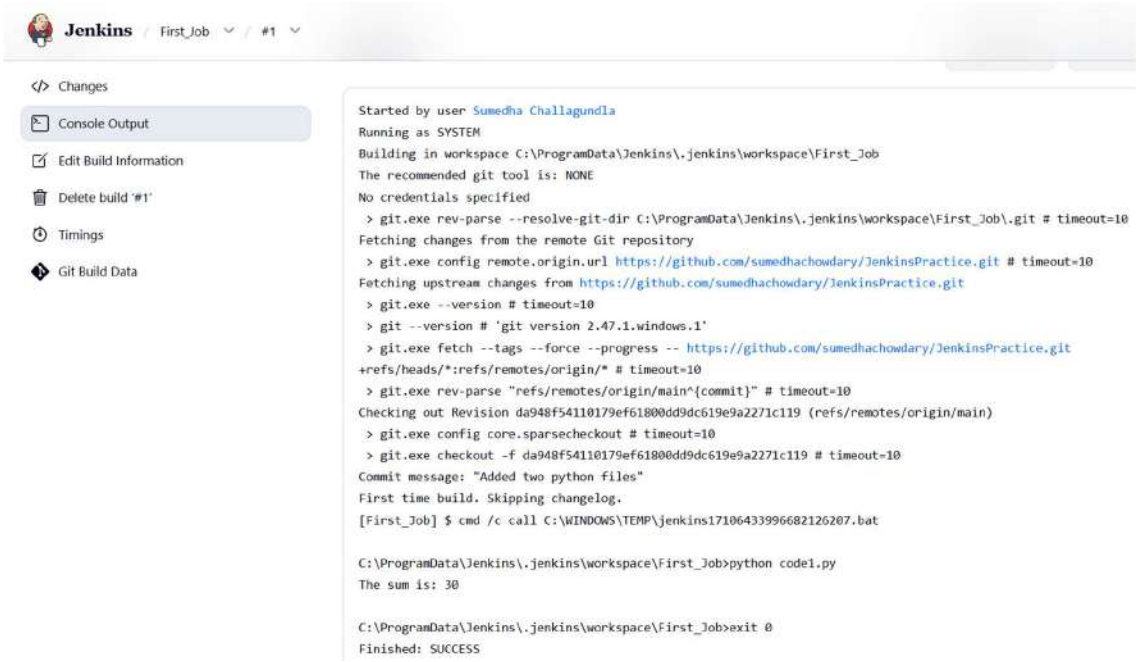
Build


Jenkins / First_Job
Status
Changes
Workspace
Build Now
Configure
Delete Project
Git Polling Log
Rename
Builds
Filter
/
Today
#1 10:52 PM
First_Job
Permalinks
Last build (#1), 22 sec ago
Last stable build (#1), 22 sec ago
Last successful build (#1), 22 sec ago
Last completed build (#1), 22 sec ago

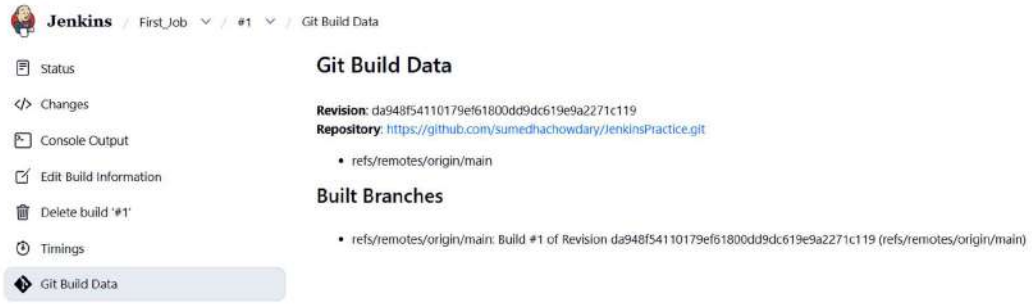
Status



Console Output



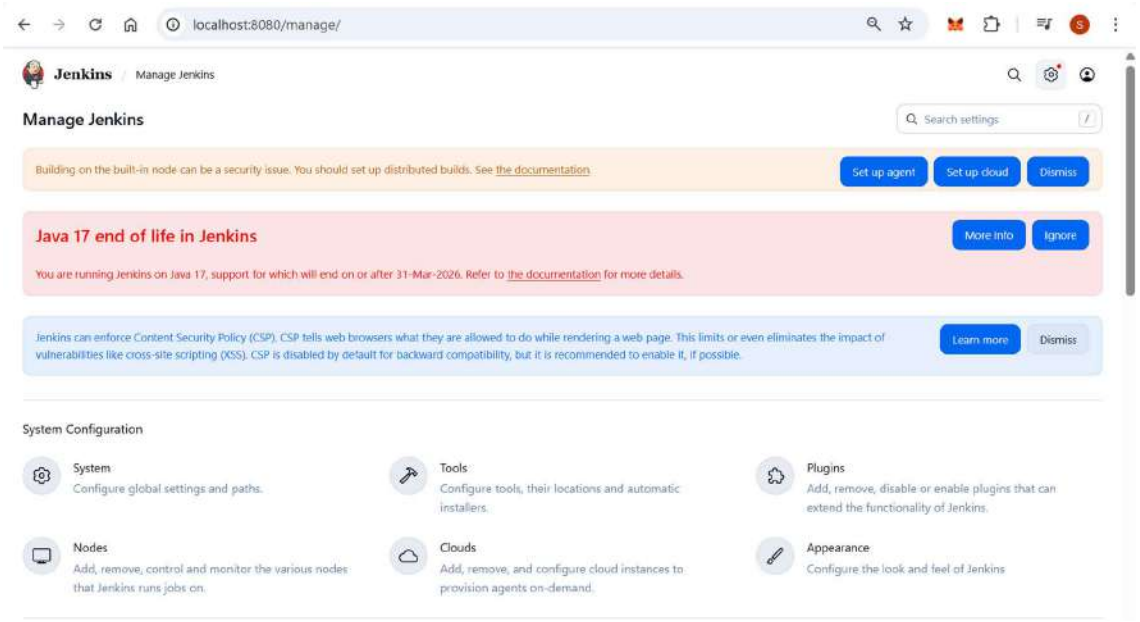
Git build data



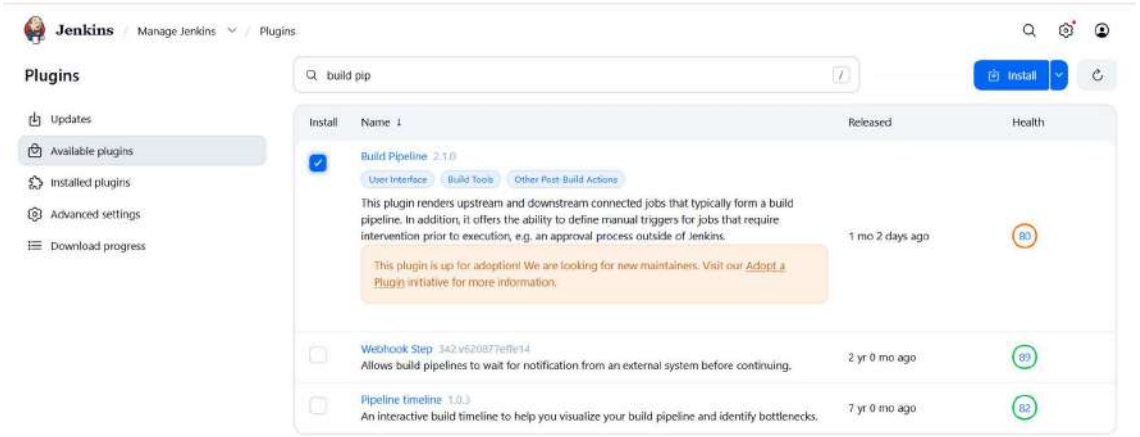
23rd jan, 2026 Lab

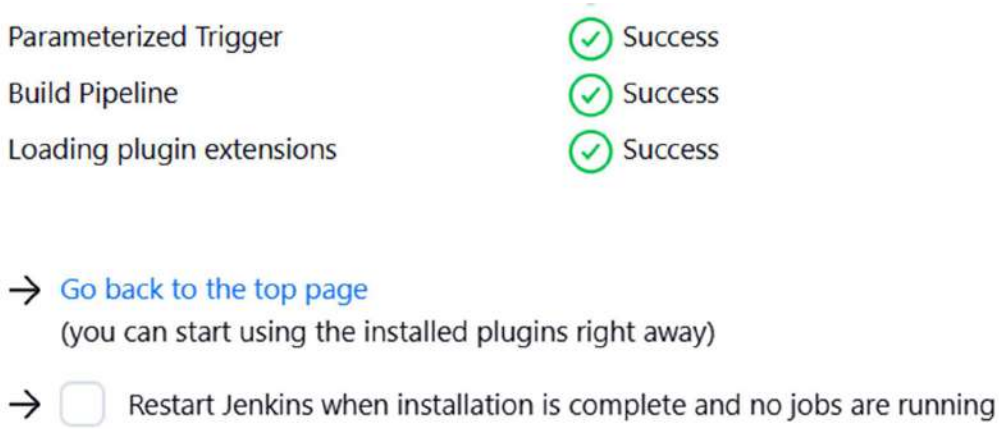
2.Creating Pipeline using Freestyle project

Manage Jenkins

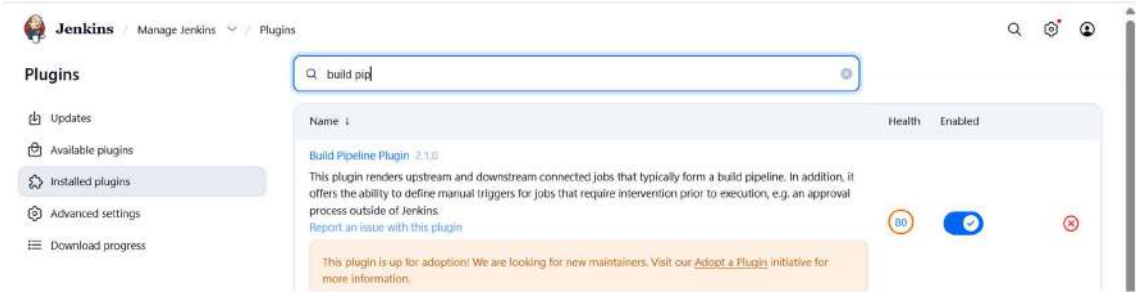


Install build pipeline plugin

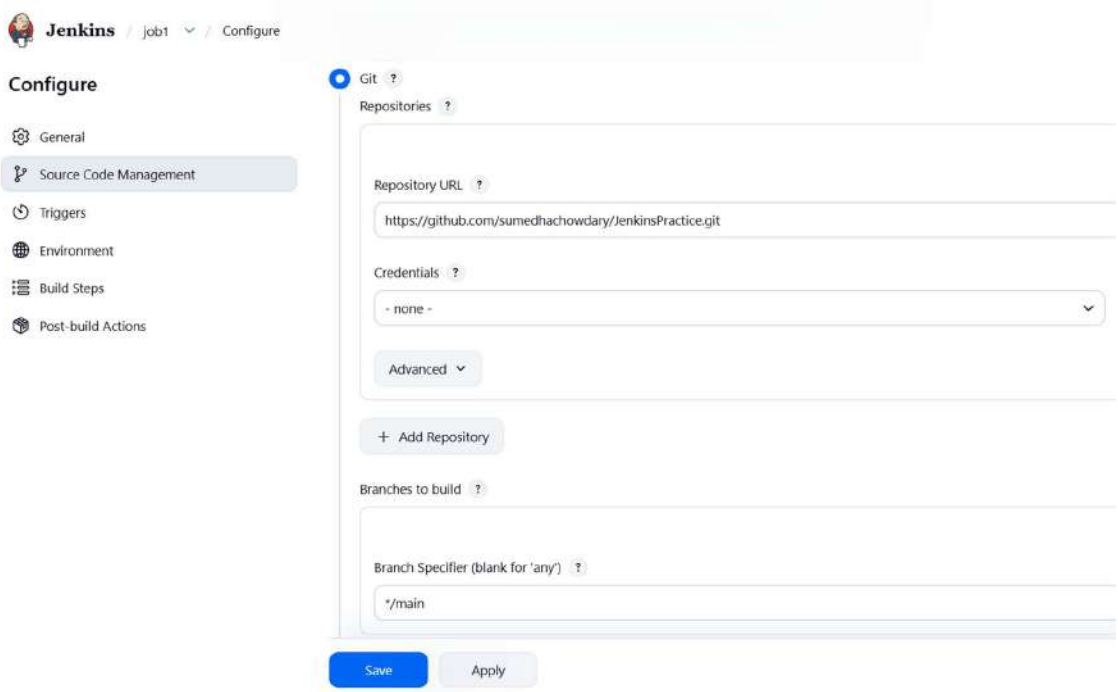




Build pipeline was installed successfully



3.Creating first job -Job1



Jenkins / job1 / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

11:27:00 pm India Standard Time.

[ ] Ignore post-commit hooks ?

Environment

Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.

[x] Delete workspace before build starts

Advanced

[ ] Use secret text(s) or file(s) ?

[ ] Add timestamps to the Console Output

[ ] Inspect build log for published build scans

[ ] Terminate a build if it's stuck

[ ] With Ant ?

Jenkins / job1

- Status
- Changes
- Workspace
- Build Now
- Configure
- Delete Project
- Git Polling Log
- Rename

Builds >

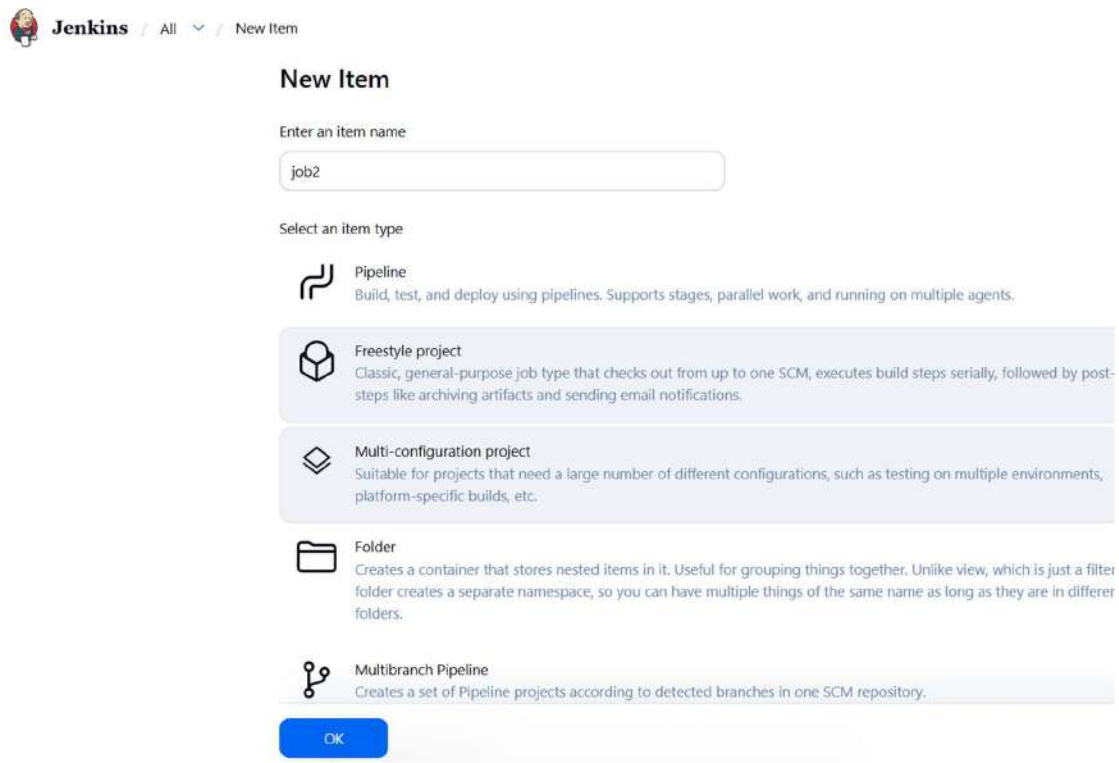
Today

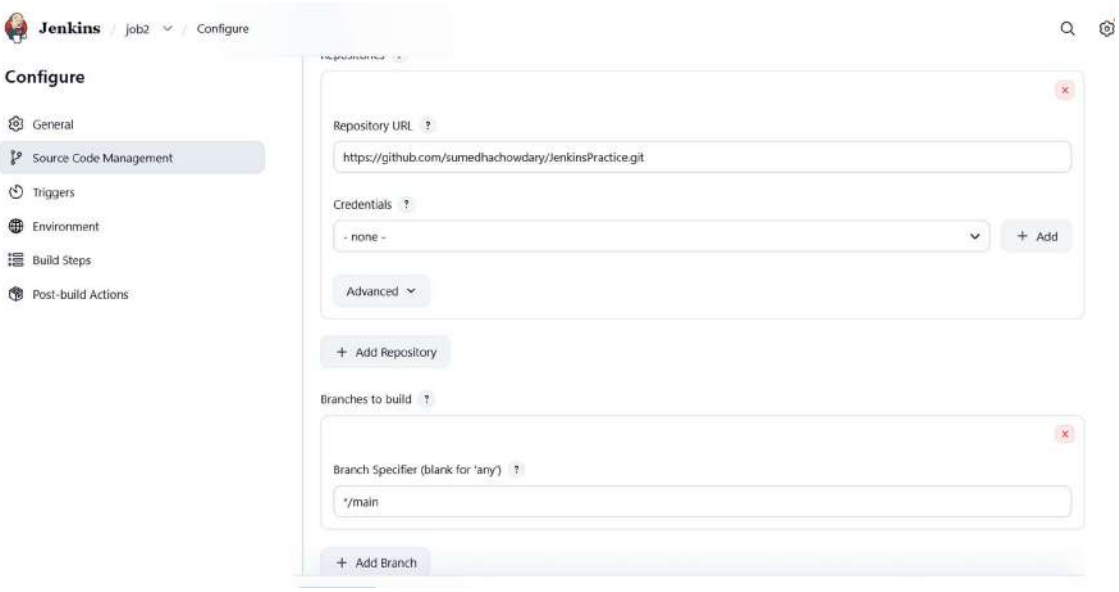
#1 11:28 PM

job1

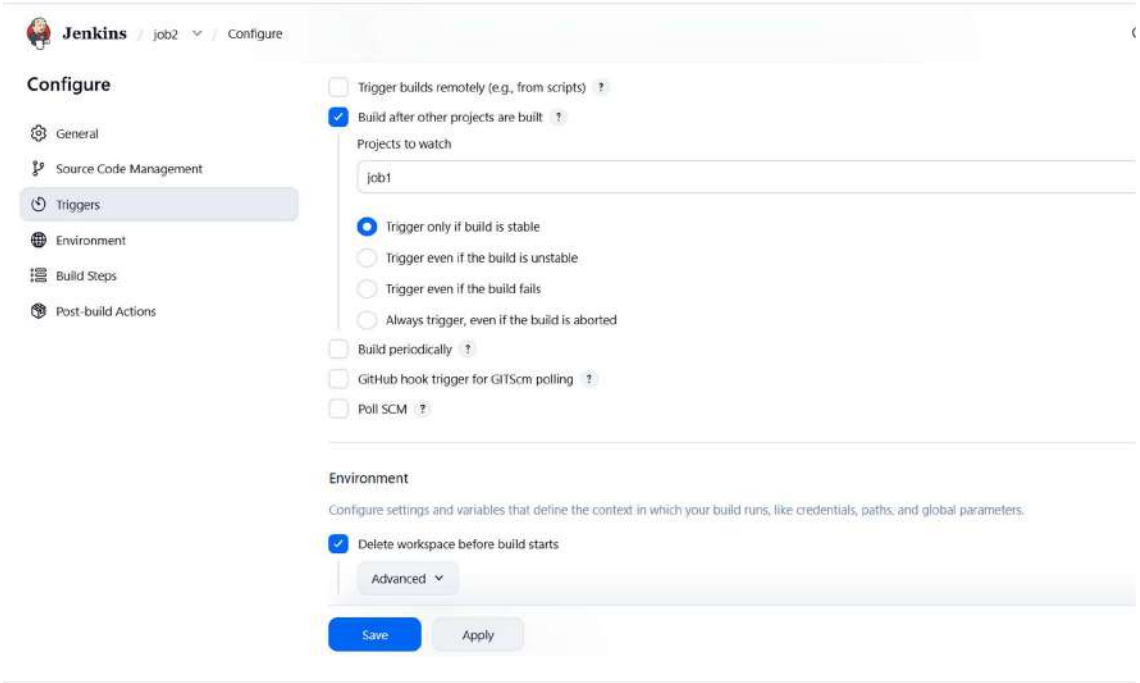
Permalinks

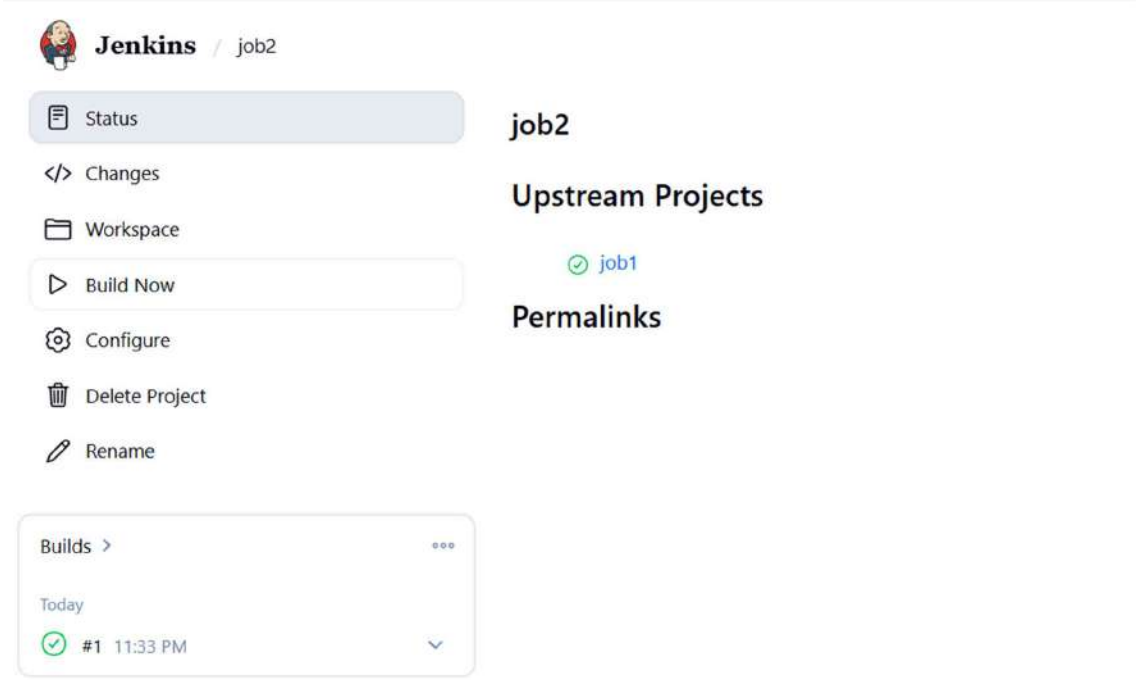
4.Creating second job-job2



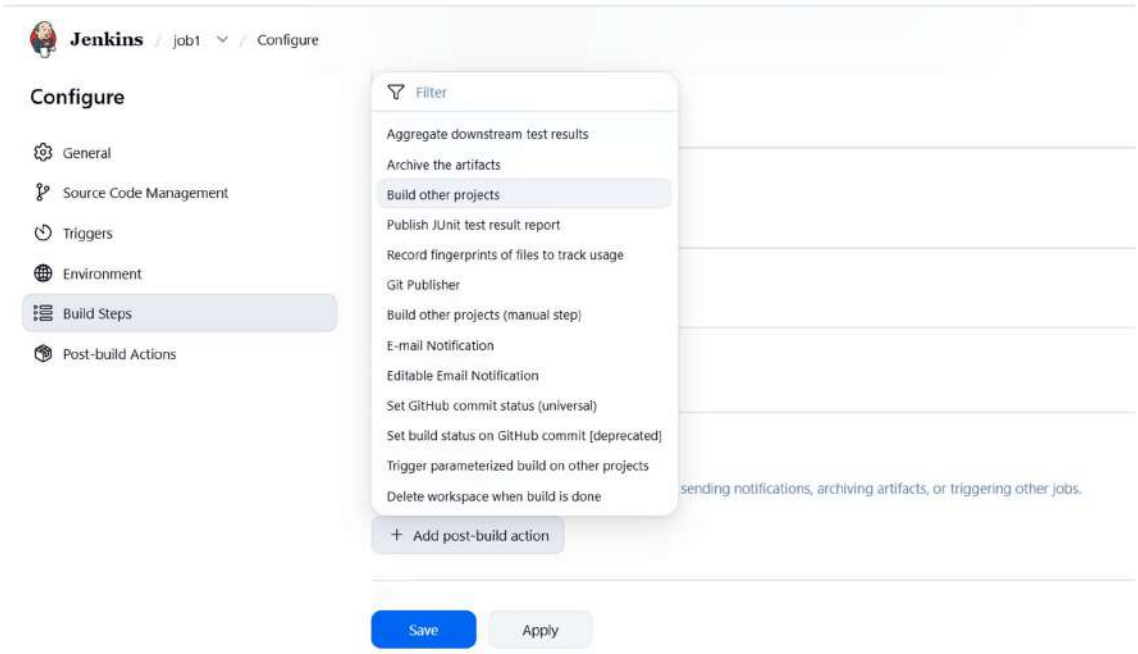


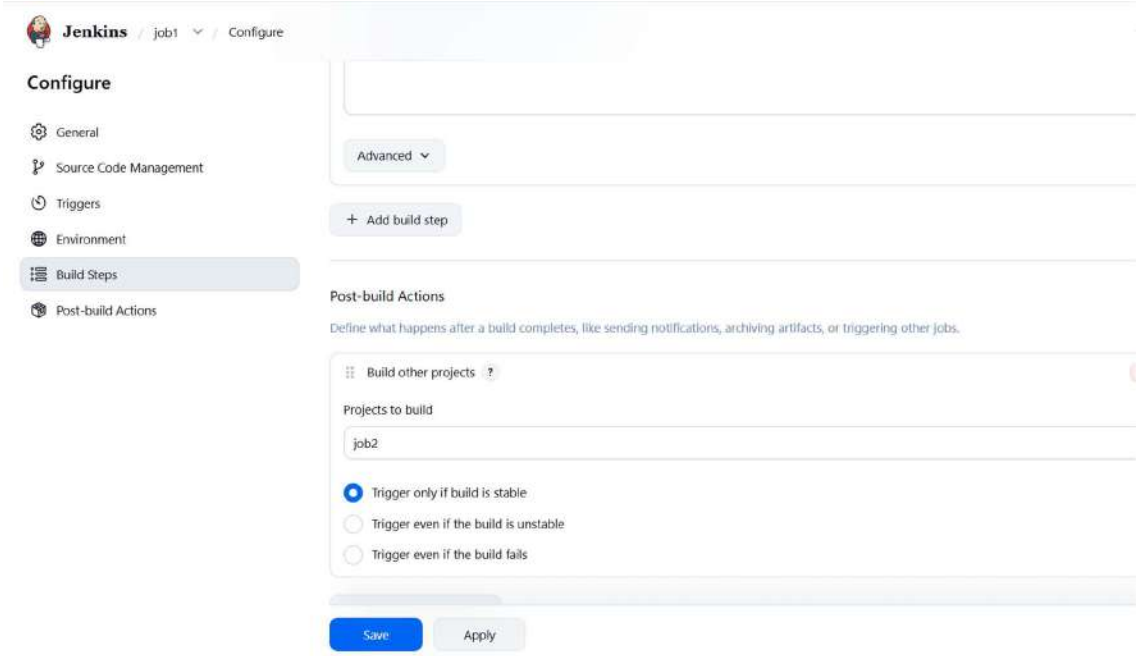
Trigger and Environment



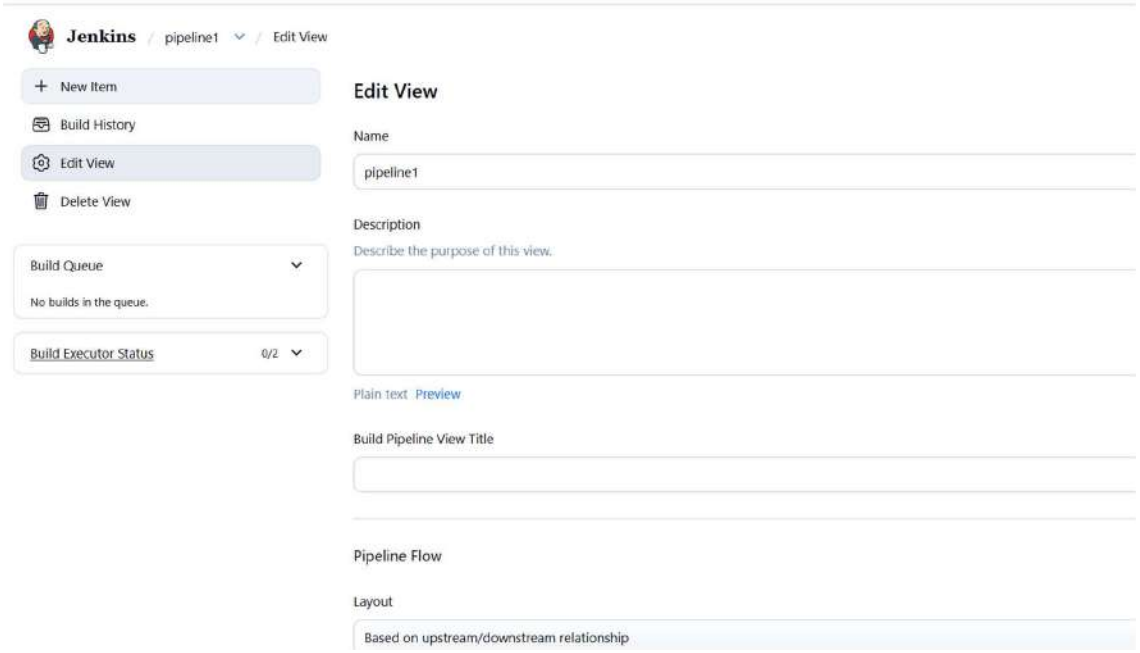


Job1 post build actions

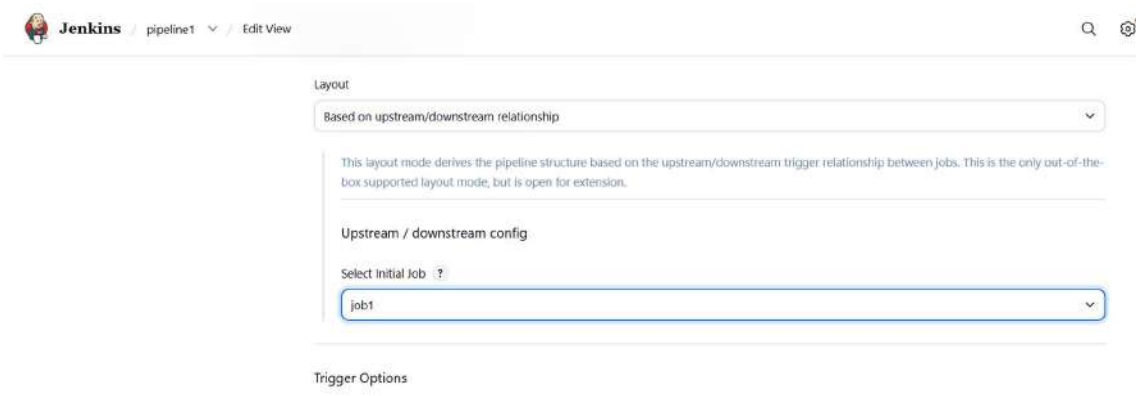


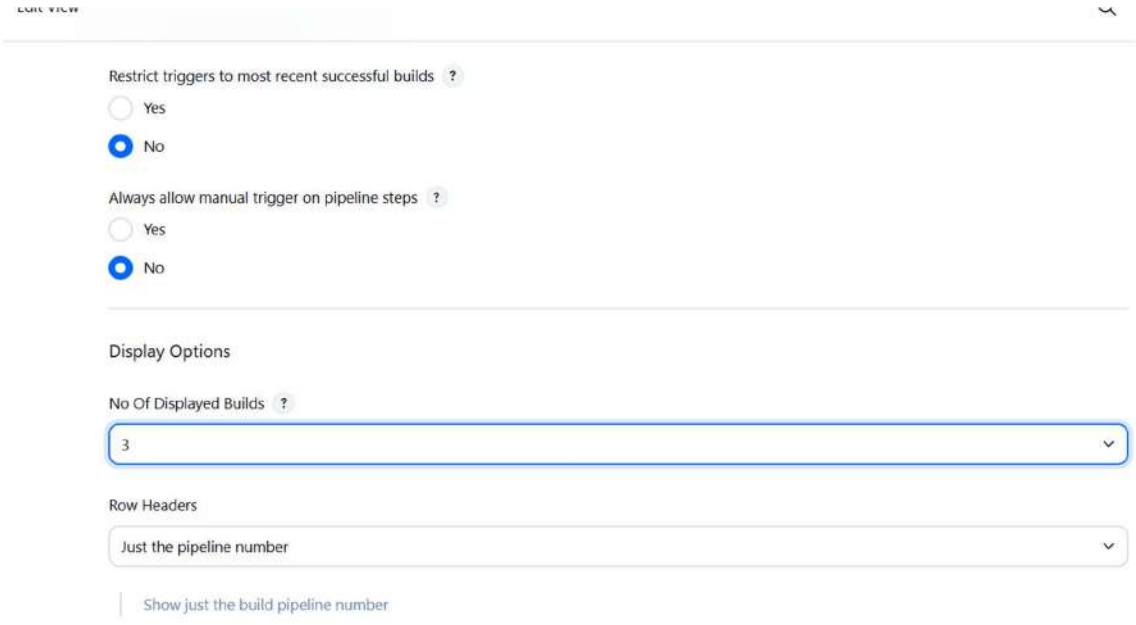


5.Create a new pipeline

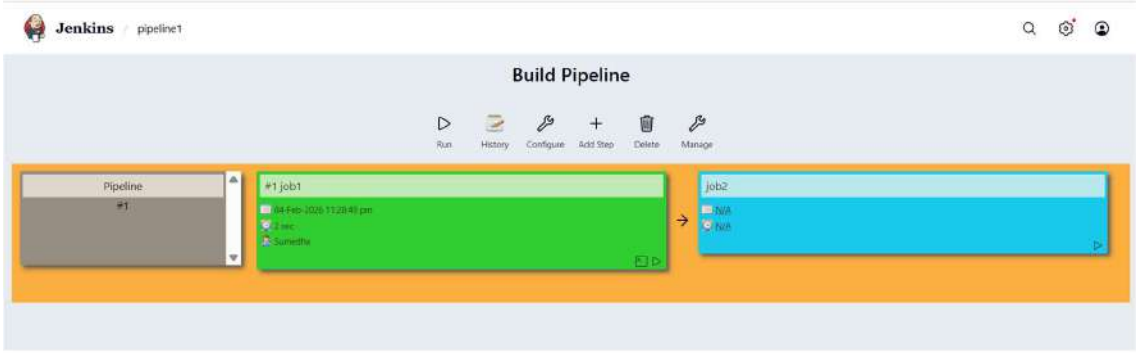


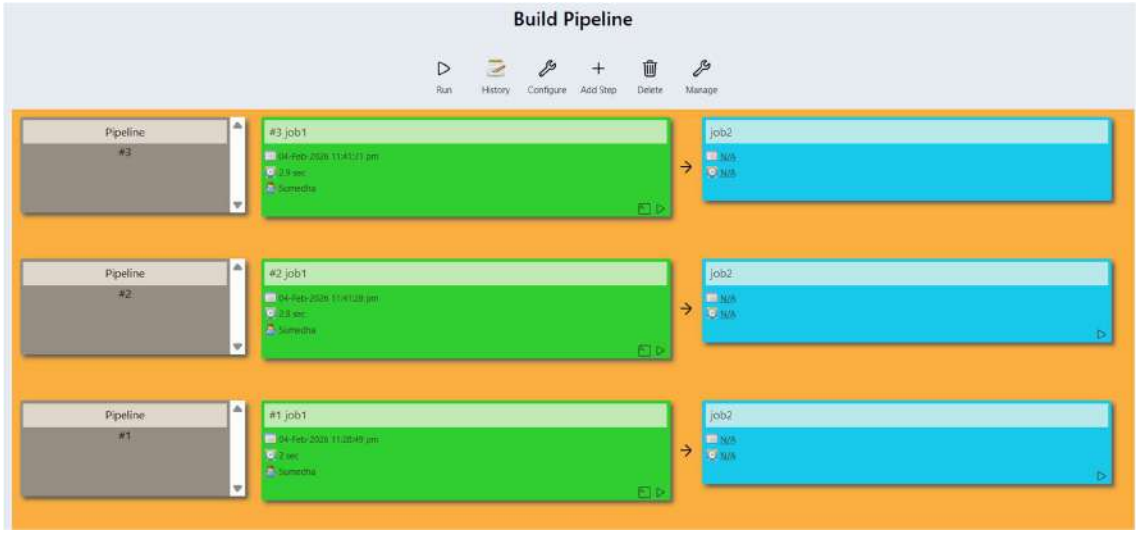
Set initial job to job1





Pipeline created

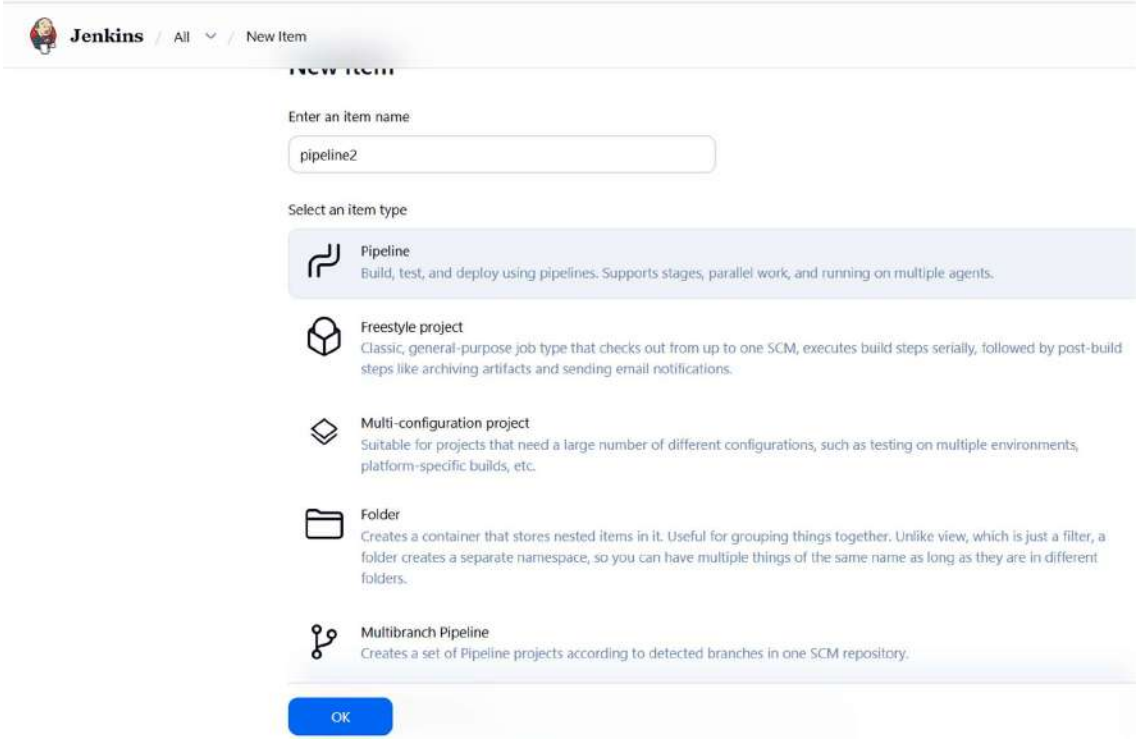




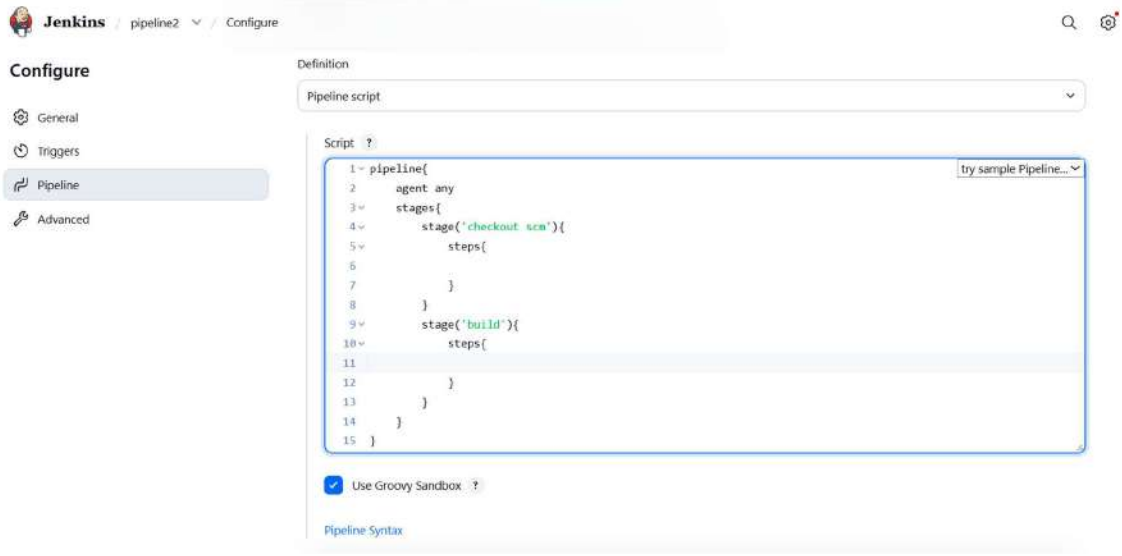
24th jan, 2026 Lab

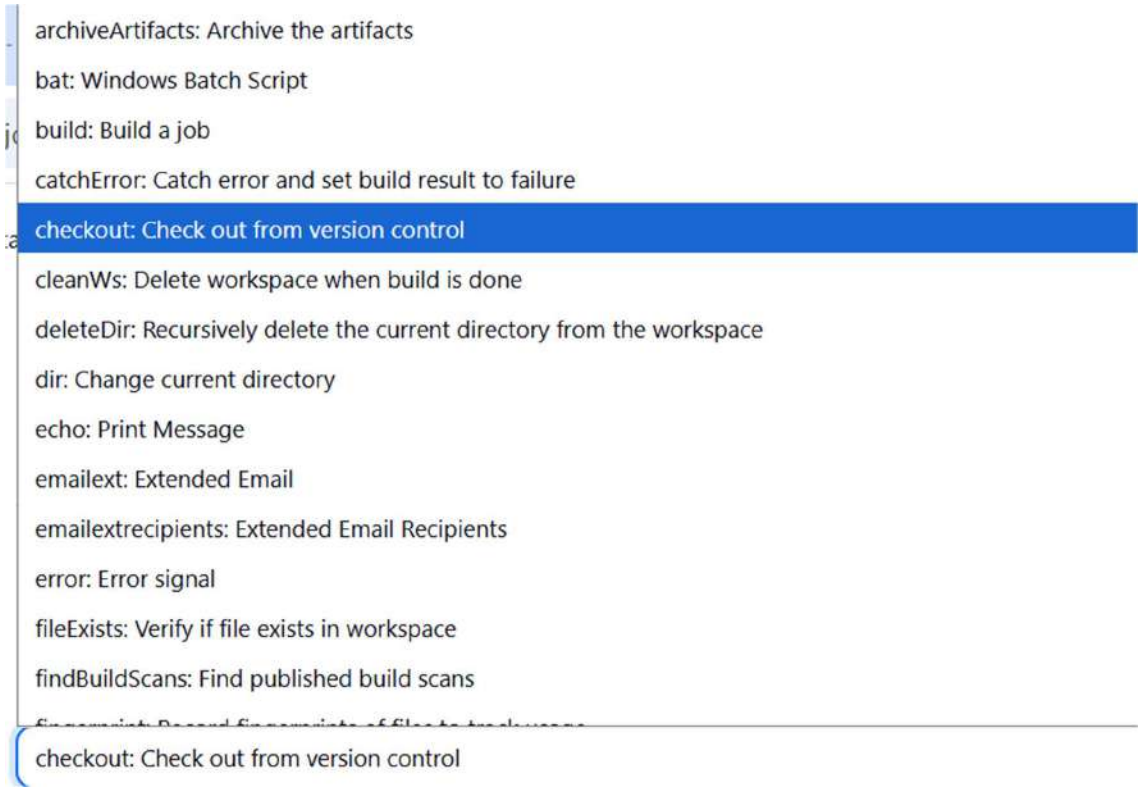
6.Method 2 of creating a pipeline

New item -> pipeline name -> pipeline

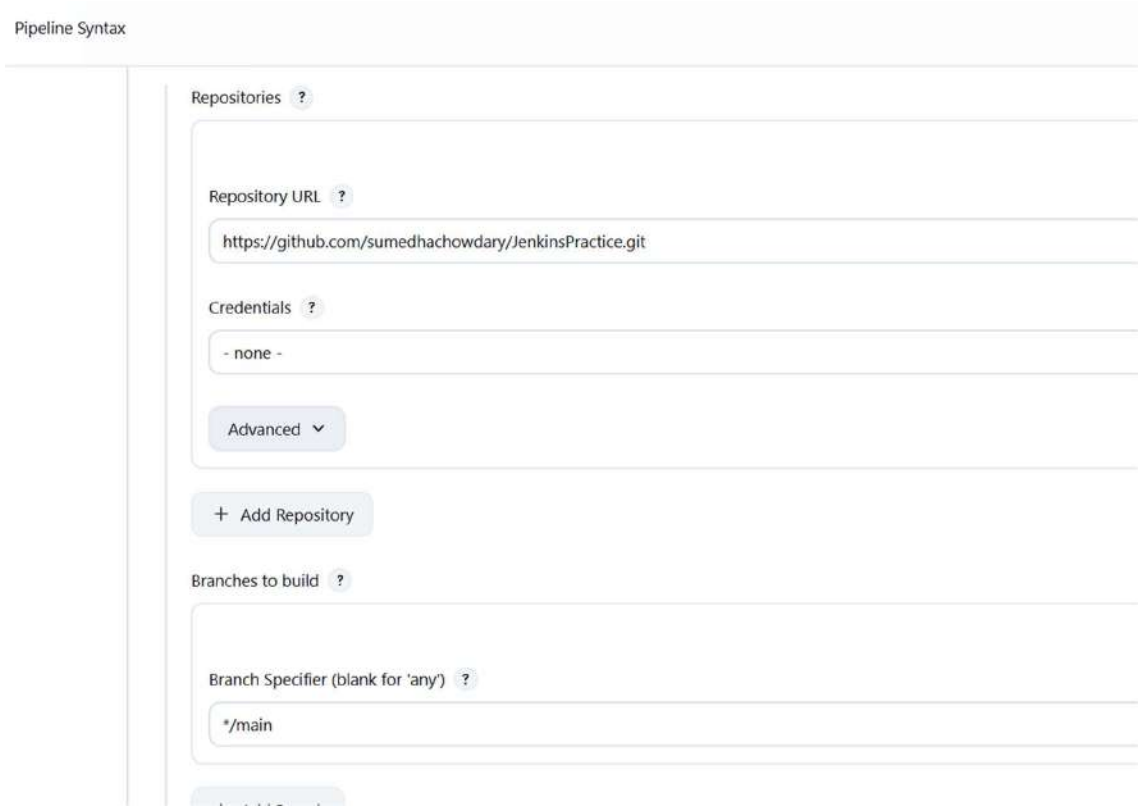


Pipeline syntax



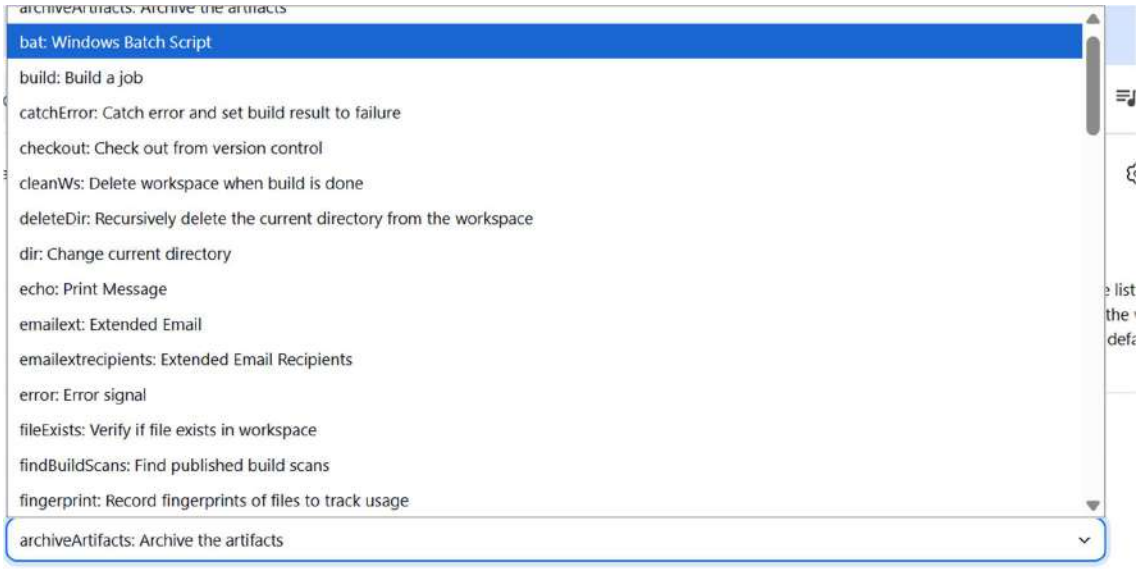


Give repository name and change branch specifier to */main



Click on generate pipeline script

Copy that pipeline script and paste it in the script



Sample Step

bat: Windows Batch Script

bat

Batch Script ?

```
dir
```

Advanced

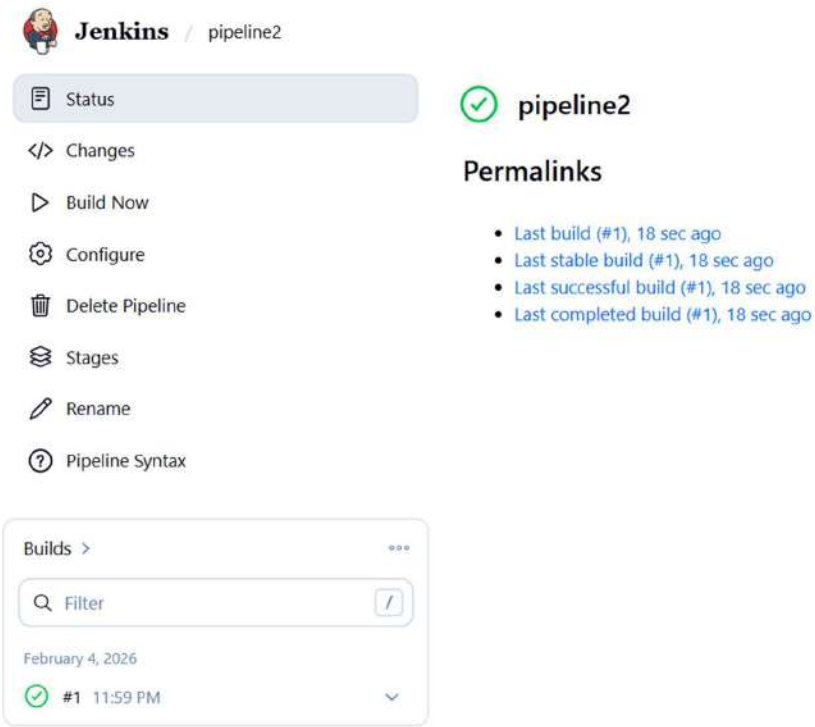
Generate Pipeline Script

```
bat 'dir'
```

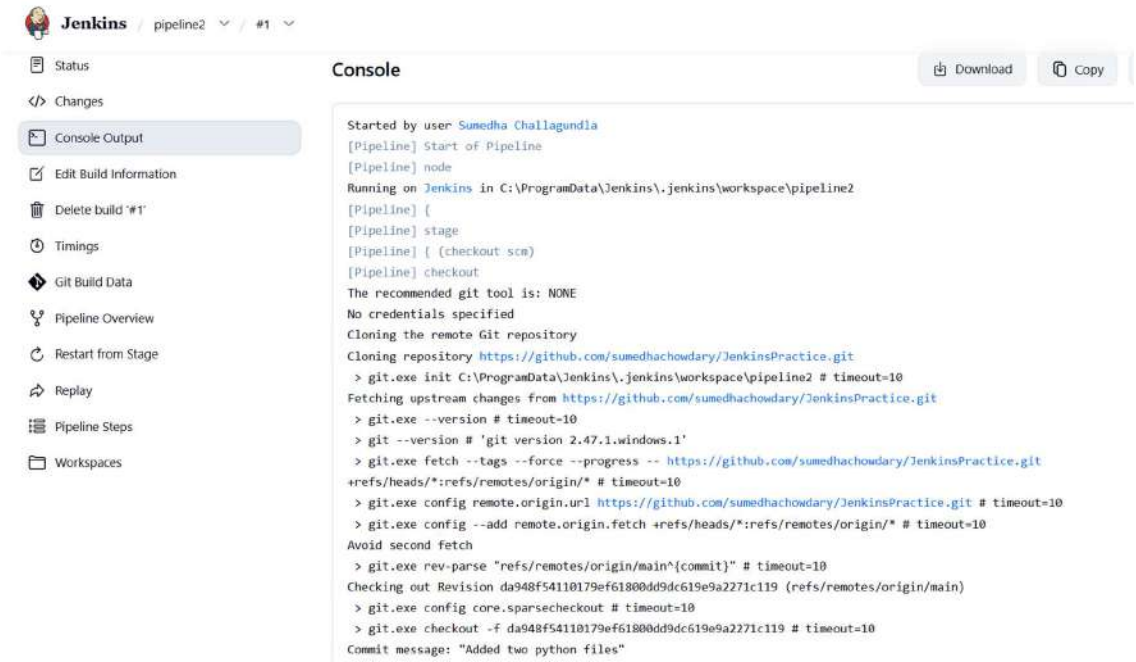
try sample Pipeline...

```
pipeline{
    agent any
    stages{
        stage('checkout scm'){
            steps{
                checkout scmGit(branches: [[name: '*/main']], extensions: [], userRemoteConfigs: [[url: 'ht
            }
        }
        stage('build'){
            steps{
                bat 'dir'
            }
        }
    }
```

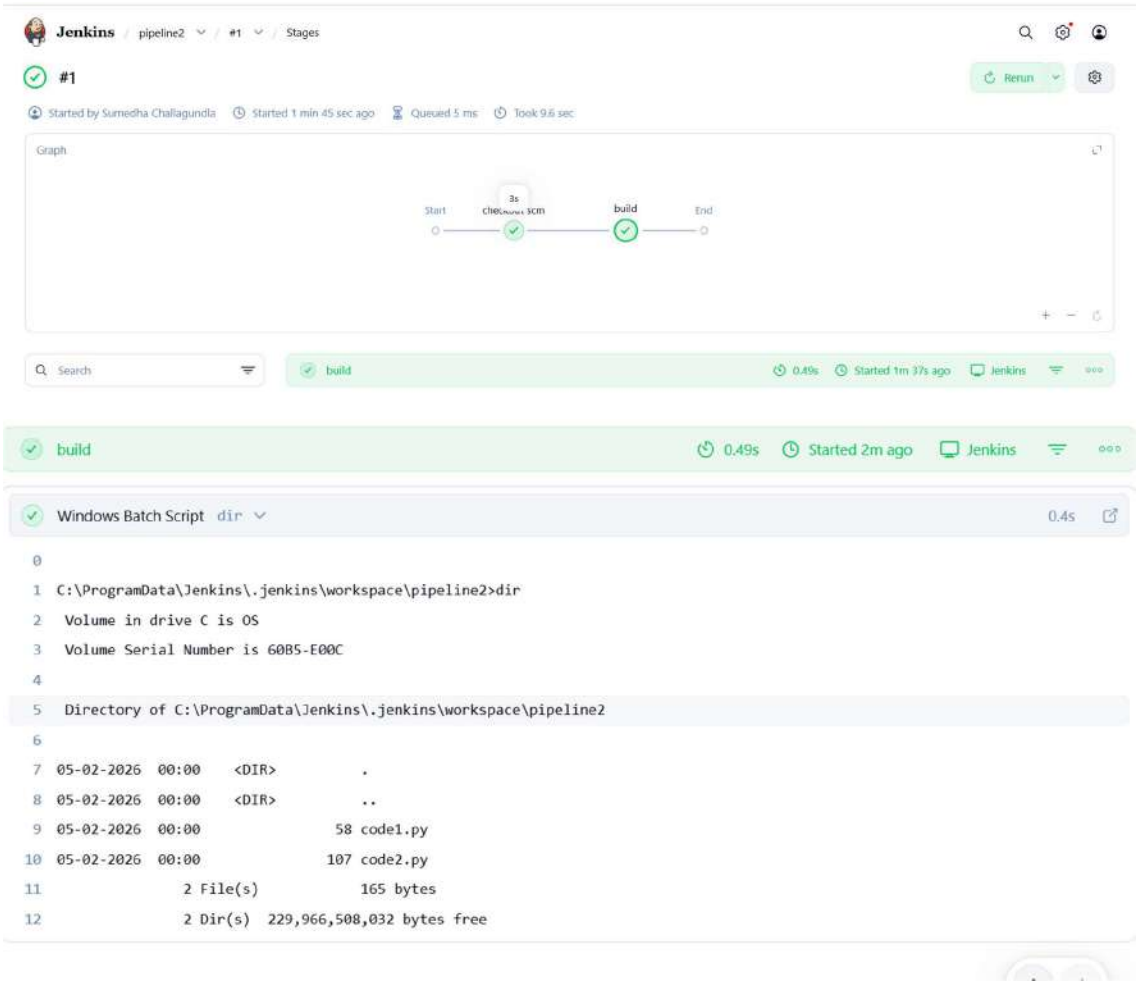
Pipeline2 is built successfully



Console output



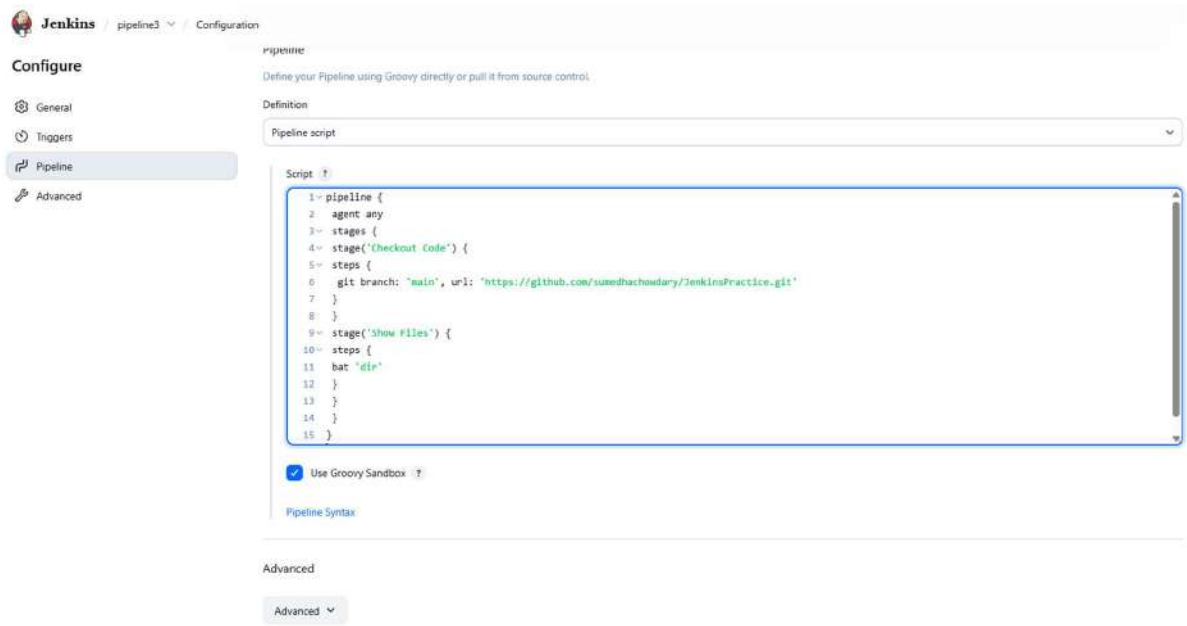
Pipeline overview



6th February, 2026

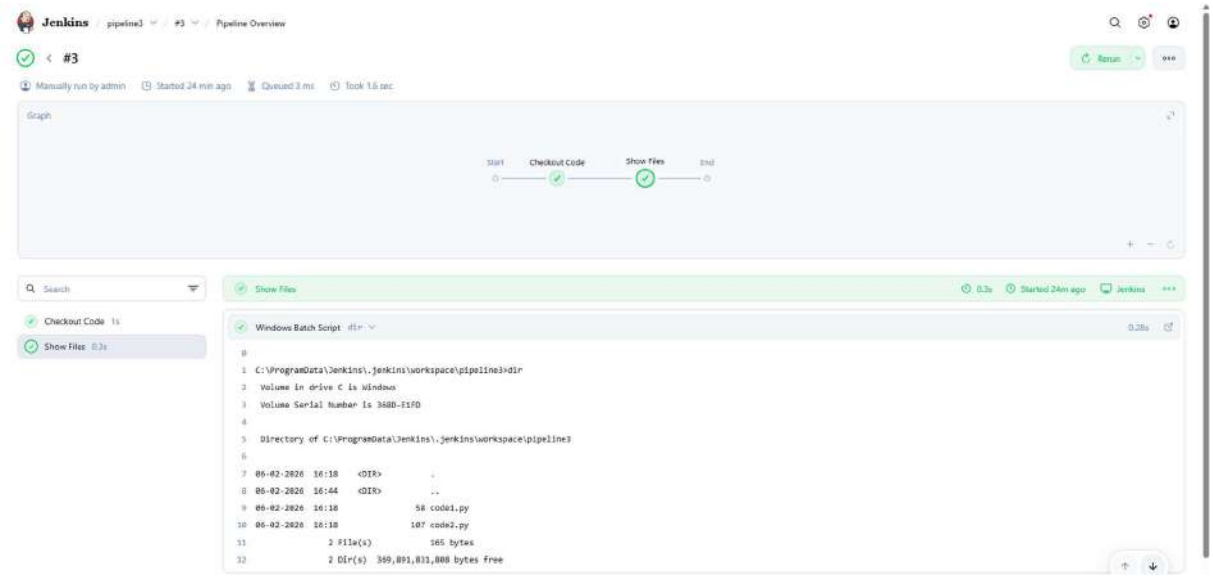
Pipeline Task 1:

pipeline script



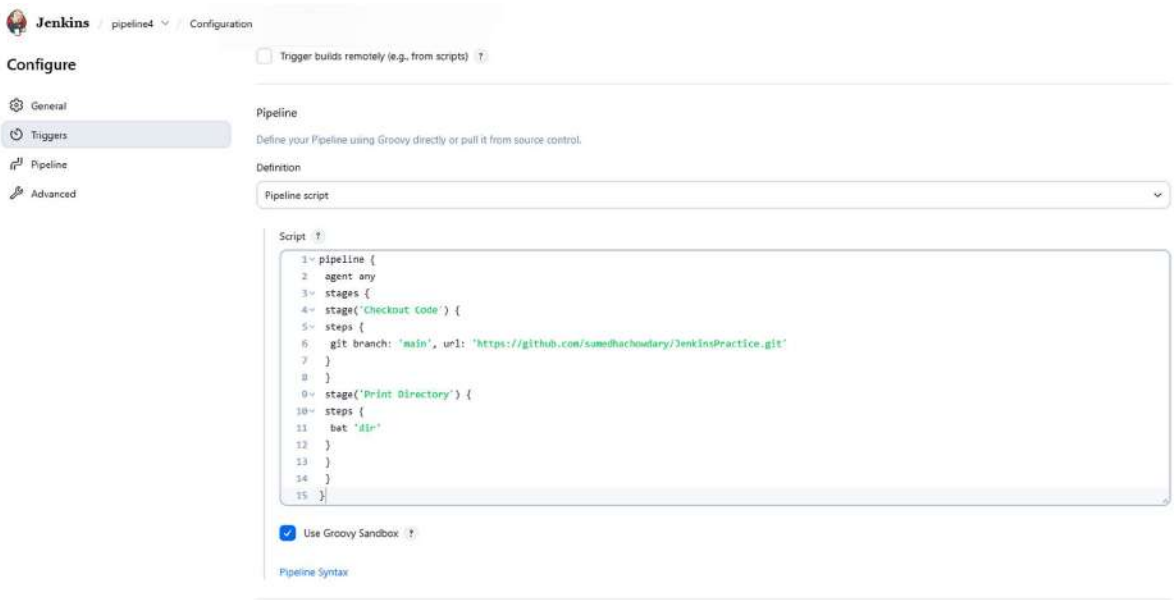
Console output

Pipeline overview



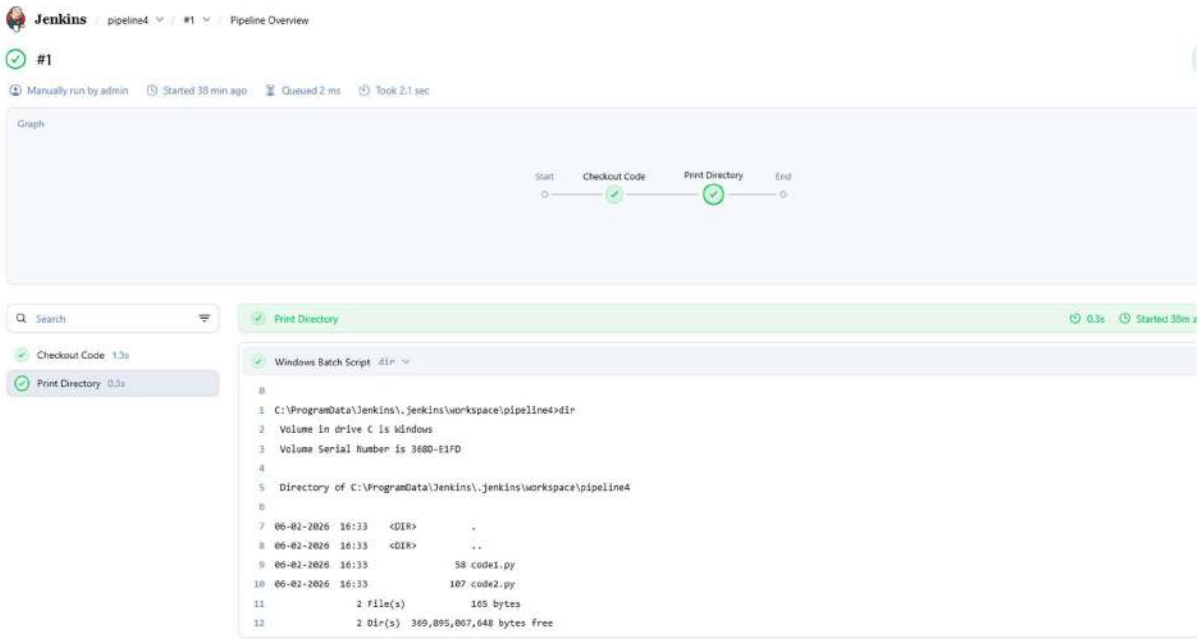
Pipeline Task 2:

Pipeline script



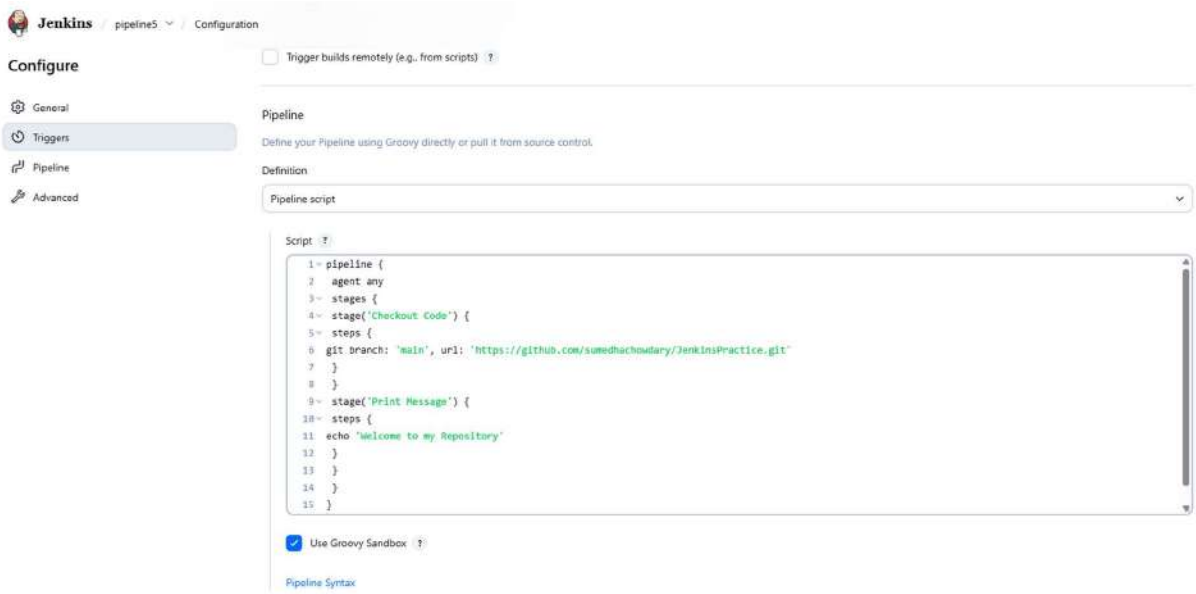
Console output

Pipeline overview

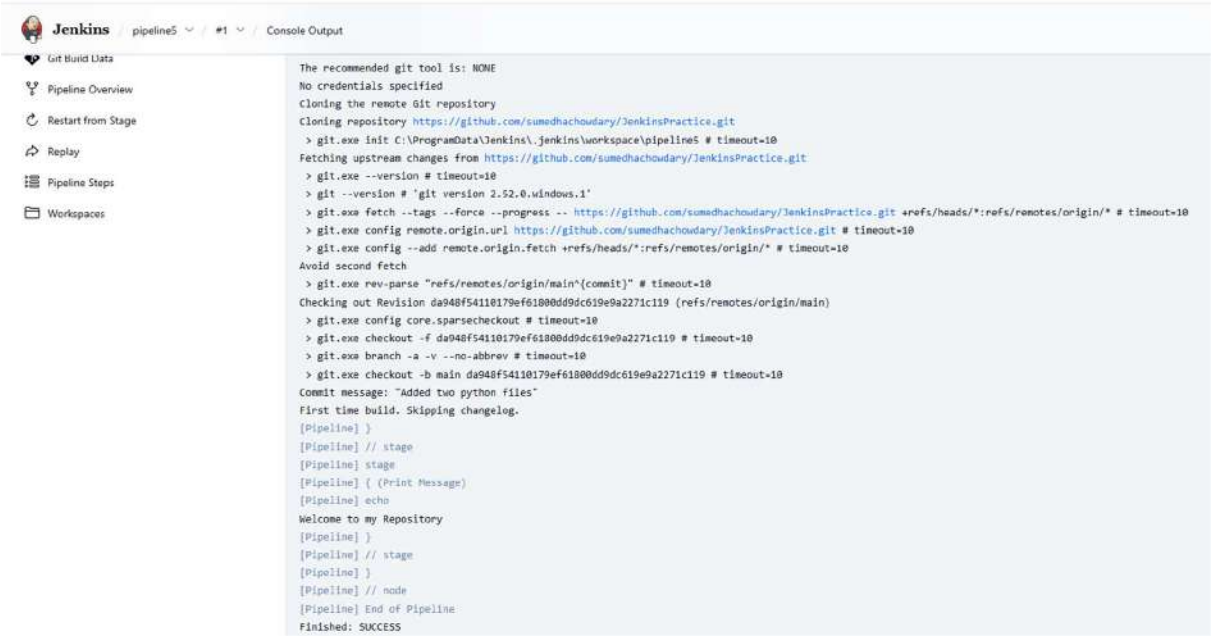


Pipeline Task 3:

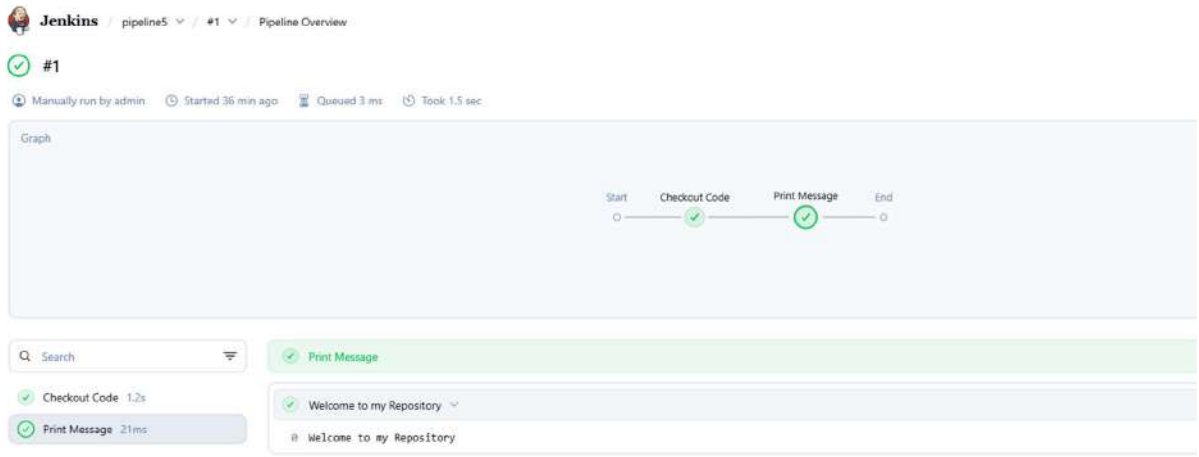
Pipeline script



Console output

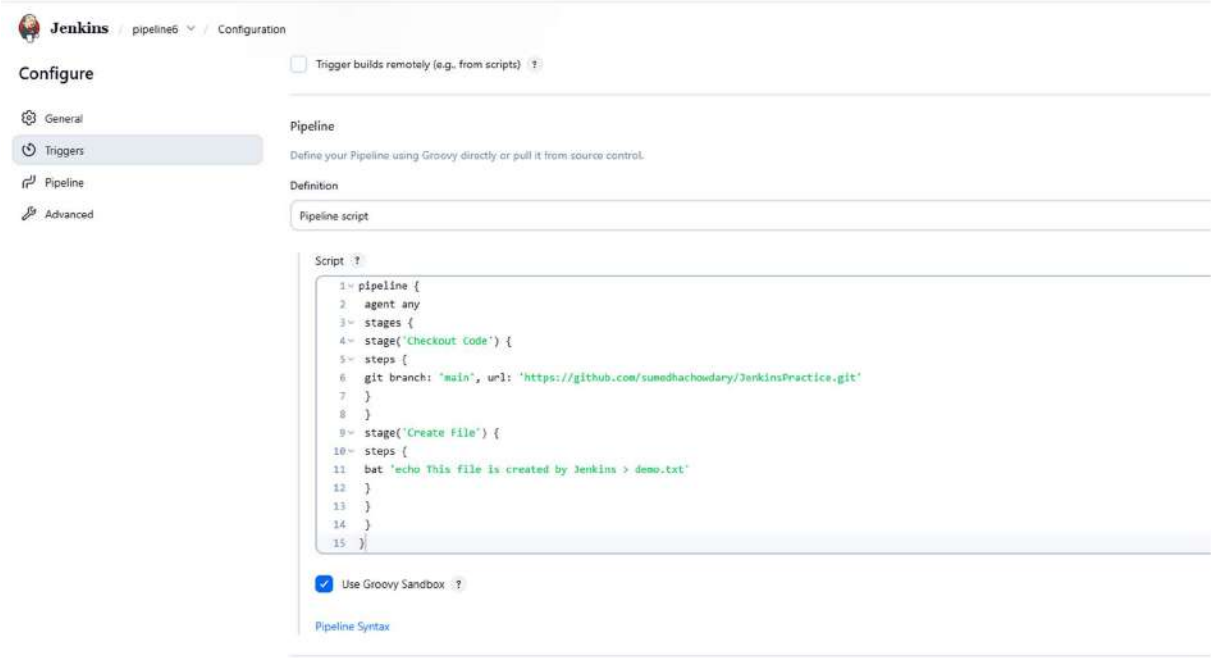


Pipeline overview

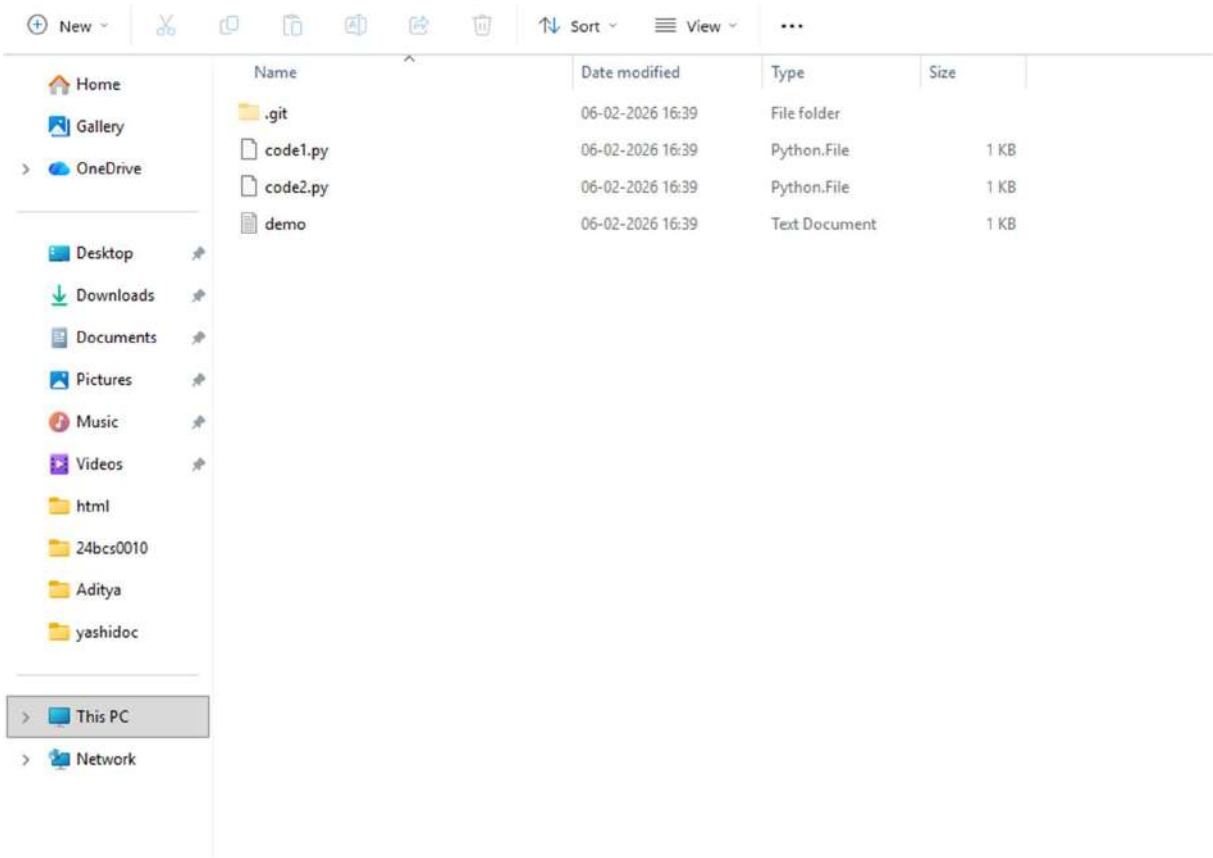


Pipeline Task 4:

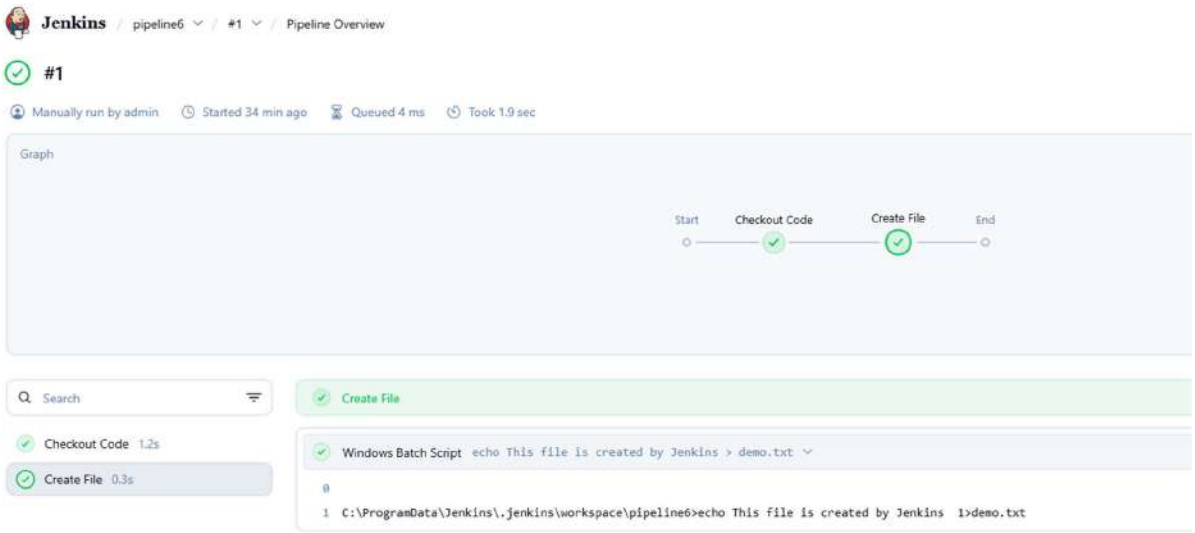
Pipeline script



Console output

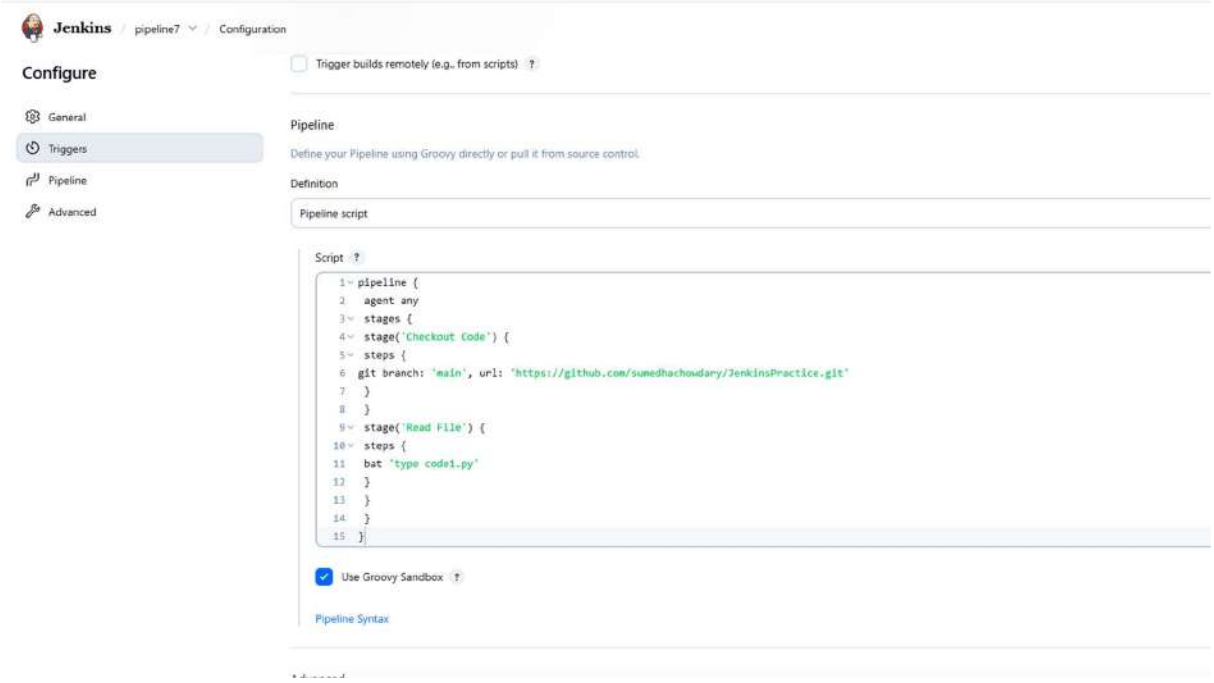


Pipeline overview



Pipeline Task 5:

Pipeline script



Console output

Jenkins / pipeline7 / #2 / Console Output

Pipeline Steps

Workspaces

```
 > git.exe --version # timeout=10
 > git --version # 'git version 2.52.0.windows.1'
 > git.exe fetch --tags --force --progress -- https://github.com/sumedhachowdary/JenkinsPractice.git +refs/heads/*:refs/remotes/origin/* # timeou
 > git.exe config remote.origin.url https://github.com/sumedhachowdary/JenkinsPractice.git # timeout=10
 > git.exe config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
 > git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision da948f54110179ef61800dd9dc619e9a2271c119 (refs/remotes/origin/main)
 > git.exe config core.sparsecheckout # timeout=10
 > git.exe checkout -f da948f54110179ef61800dd9dc619e9a2271c119 # timeout=10
 > git.exe branch -a -v --no-abbrev # timeout=10
 > git.exe checkout -b main da948f54110179ef61800dd9dc619e9a2271c119 # timeout=10
Commit message: "Added two python files"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Read File)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\pipeline7>type code1.py
a = 10
b = 20
sum = a + b

print("The sum is:", sum)
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Pipeline overview

Jenkins / pipeline7 / #2 / Pipeline Overview

#2

Manually run by admin Started 29 min ago Queued 3 ms Took 2.4 sec

Graph

Start → Checkout Code → Read File → End

Search

Checkout Code 1.8s

Read File 0.32s

Read File

Windows Batch Script type code1.py

```
0
1 C:\ProgramData\Jenkins\.jenkins\workspace\pipeline7>type code1.py
2 a = 10
3 b = 20
4 sum = a + b
5
6 print("The sum is:", sum)
```

THE END